

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent
Kind Code
Date of Patent
Inventor(s)

12408291
B2
September 02, 2025
Yeung; Tony et al.

Enclosure assembly for enhanced cooling of direct drive unit and related methods

Abstract

Embodiments of an enclosure assembly to enhance cooling of a hydraulic fracturing direct drive unit (DDU) during operation are included. The enclosure assembly may include an enclosure body extending at least partially around an enclosure space to house the DDU for driving a fluid pump. The enclosure assembly may include one or more heat exchanger assemblies connected to the enclosure body for cooling a process fluid associated with one or more of the DDU and the fluid pump, and which may be configured to draw air into the enclosure space from an external environment, toward one or more radiator assemblies to cool the process fluid, and along an airflow path through the enclosure space. One or more outlet fan assemblies may be operative to discharge air from the enclosure space to the external environment to maintain a desired temperature of the enclosure space.

Inventors: Yeung; Tony (Houston, TX), Rodriguez-Ramon; Ricardo (Houston, TX), Rodriguez; Guillermo (Houston, TX)

Applicant: BJ Energy Solutions, LLC (Houston, TX)

Family ID: 77465198

Assignee: BJ Energy Solutions, LLC (The Woodlands, TX)

Appl. No.: 18/327599

Filed: June 01, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230320025 A1	Oct. 05, 2023

Related U.S. Application Data

continuation parent-doc US 18096927 20230113 US 11723171 child-doc US 18327599
continuation parent-doc US 17444485 20210805 US 11627683 20230411 child-doc US 18096927
continuation parent-doc US 17356063 20210623 US 11129295 20210921 child-doc US 17444485
division parent-doc US 17302039 20210422 US 11109508 20210831 child-doc US 17356063
us-provisional-application US 62705042 20200609
us-provisional-application US 62704981 20200605

Publication Classification

Int. Cl.: H05K7/20 (20060101); E21B43/26 (20060101)

U.S. Cl.:

CPC H05K7/20154 (20130101); E21B43/26 (20130101); H05K7/20145 (20130101); H05K7/20209 (20130101);

Field of Classification Search

CPC: H05K (7/20145); H05K (7/20154); F16N (7/00); H02K (7/1823); H02K (9/06); F04B (17/05); F04B (2203/11); F04B (51/00); F04B (15/02); E21B (43/2607)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1716049	12/1928	Greve	N/A	N/A
1726633	12/1928	Smith	N/A	N/A

2178662	12/1938	Lars	N/A	N/A
2427638	12/1946	Vilter	N/A	N/A
2498229	12/1949	Adler	N/A	N/A
2535703	12/1949	Smith et al.	N/A	N/A
2572711	12/1950	Fischer	N/A	N/A
2820341	12/1957	Amann	N/A	N/A
2868004	12/1958	Runde	N/A	N/A
2940377	12/1959	Darnell et al.	N/A	N/A
2947141	12/1959	Russ	N/A	N/A
2956738	12/1959	Rosenschold	N/A	N/A
3068796	12/1961	Pfluger et al.	N/A	N/A
3191517	12/1964	Solzman	N/A	N/A
3257031	12/1965	Dietz	N/A	N/A
3274768	12/1965	Klein	N/A	N/A
3378074	12/1967	Kiel	N/A	N/A
3382671	12/1967	Ehni, III	N/A	N/A
3401873	12/1967	Privon	N/A	N/A
3463612	12/1968	Whitsel	N/A	N/A
3496880	12/1969	Wolff	N/A	N/A
3550696	12/1969	Kenneday	N/A	N/A
3560053	12/1970	Ortloff	N/A	N/A
3586459	12/1970	Zerlauth	N/A	N/A
3632222	12/1971	Cronstedt	N/A	N/A
3656582	12/1971	Alcock	N/A	N/A
3667868	12/1971	Brunner	N/A	N/A
3692434	12/1971	Schnear	N/A	N/A
3739872	12/1972	McNair	N/A	N/A
3757581	12/1972	Mankin	N/A	N/A
3759063	12/1972	Bendall	N/A	N/A
3765173	12/1972	Harris	N/A	N/A
3771916	12/1972	Flanigan et al.	N/A	N/A
3773438	12/1972	Hall et al.	N/A	N/A
3781135	12/1972	Nickell	N/A	N/A
3786835	12/1973	Finger	N/A	N/A
3791682	12/1973	Mitchell	N/A	N/A
3796045	12/1973	Foster	N/A	N/A
3814549	12/1973	Cronstedt	N/A	N/A
3820922	12/1973	Buse et al.	N/A	N/A
3847511	12/1973	Cole	N/A	N/A
3866108	12/1974	Yannone	N/A	N/A
3875380	12/1974	Rankin	N/A	N/A
3963372	12/1975	McLain et al.	N/A	N/A
4010613	12/1976	McInerney	N/A	N/A
4019477	12/1976	Overton	N/A	N/A
4031407	12/1976	Reed	N/A	N/A
4050862	12/1976	Buse	N/A	N/A
4059045	12/1976	McClain	N/A	N/A
4086976	12/1977	Holm et al.	N/A	N/A
4117342	12/1977	Melley, Jr.	N/A	N/A
4173121	12/1978	Yu	N/A	N/A
4204808	12/1979	Reese et al.	N/A	N/A
4209079	12/1979	Marchal et al.	N/A	N/A
4209979	12/1979	Woodhouse et al.	N/A	N/A
4222229	12/1979	Uram	N/A	N/A
4239396	12/1979	Arribau et al.	N/A	N/A
4269569	12/1980	Hoover	N/A	N/A
4311395	12/1981	Douthitt et al.	N/A	N/A
4330237	12/1981	Battah	N/A	N/A
4341508	12/1981	Rambin, Jr.	N/A	N/A
4357027	12/1981	Zeitlow	N/A	N/A
4383478	12/1982	Jones	N/A	N/A
4402504	12/1982	Christian	N/A	N/A
4430047	12/1983	Ilg	N/A	N/A
4442665	12/1983	Fick	N/A	N/A
4457325	12/1983	Green	N/A	N/A
4470771	12/1983	Hall et al.	N/A	N/A
4483684	12/1983	Black	N/A	N/A
4505650	12/1984	Hannett et al.	N/A	N/A
4574880	12/1985	Handke	N/A	N/A
4584654	12/1985	Crane	N/A	N/A
4620330	12/1985	Izzi, Sr.	N/A	N/A

4672813	12/1986	David	N/A	N/A
4754607	12/1987	Mackay	N/A	N/A
4782244	12/1987	Wakimoto	N/A	N/A
4796777	12/1988	Keller	N/A	N/A
4869209	12/1988	Young	N/A	N/A
4913625	12/1989	Gerlowski	N/A	N/A
4983259	12/1990	Duncan	N/A	N/A
4990058	12/1990	Eslinger	N/A	N/A
5032065	12/1990	Yamamuro	N/A	N/A
5135361	12/1991	Dion	N/A	N/A
5167493	12/1991	Kobari	N/A	N/A
5245970	12/1992	Iwaszkiewicz et al.	N/A	N/A
5291842	12/1993	Sallstrom et al.	N/A	N/A
5326231	12/1993	Pandeya	N/A	N/A
5362219	12/1993	Paul et al.	N/A	N/A
5511956	12/1995	Hasegawa	N/A	N/A
5537813	12/1995	Davis et al.	N/A	N/A
5553514	12/1995	Walkowc	N/A	N/A
5560195	12/1995	Anderson et al.	N/A	N/A
5586444	12/1995	Fung	N/A	N/A
5622245	12/1996	Reik	N/A	N/A
5626103	12/1996	Haws et al.	N/A	N/A
5634777	12/1996	Albertin	N/A	N/A
5651400	12/1996	Corts et al.	N/A	N/A
5678460	12/1996	Walkowc	N/A	N/A
5717172	12/1997	Griffin, Jr. et al.	N/A	N/A
5720598	12/1997	de Chizzelle	N/A	N/A
5761084	12/1997	Edwards	N/A	N/A
5811676	12/1997	Spalding et al.	N/A	N/A
5839888	12/1997	Harrison	N/A	N/A
5846062	12/1997	Yanagisawa et al.	N/A	N/A
5875744	12/1998	Vallejos	N/A	N/A
5983962	12/1998	Gerardot	N/A	N/A
5992944	12/1998	Hara	N/A	N/A
6041856	12/1999	Thrasher et al.	N/A	N/A
6050080	12/1999	Horner	N/A	N/A
6067962	12/1999	Bartley et al.	N/A	N/A
6071188	12/1999	O'Neill et al.	N/A	N/A
6074170	12/1999	Bert et al.	N/A	N/A
6123751	12/1999	Nelson et al.	N/A	N/A
6129335	12/1999	Yokogi	N/A	N/A
6145318	12/1999	Kaplan et al.	N/A	N/A
6230481	12/2000	Jahr	N/A	N/A
6279309	12/2000	Lawlor, II et al.	N/A	N/A
6321860	12/2000	Reddoch	N/A	N/A
6334746	12/2001	Nguyen et al.	N/A	N/A
6401472	12/2001	Pollrich	N/A	N/A
6530224	12/2002	Conchieri	N/A	N/A
6543395	12/2002	Green	N/A	N/A
6655922	12/2002	Flek	N/A	N/A
6669453	12/2002	Breeden	N/A	N/A
6765304	12/2003	Baten et al.	N/A	N/A
6786051	12/2003	Kristich et al.	N/A	N/A
6832900	12/2003	Leu	N/A	N/A
6851514	12/2004	Han et al.	N/A	N/A
6859740	12/2004	Stephenson et al.	N/A	N/A
6901735	12/2004	Lohn	N/A	N/A
6962057	12/2004	Kurokawa et al.	N/A	N/A
7007966	12/2005	Campion	N/A	N/A
7047747	12/2005	Tanaka	N/A	N/A
7065953	12/2005	Kopko	N/A	N/A
7143016	12/2005	Discenzo et al.	N/A	N/A
7222015	12/2006	Davis et al.	N/A	N/A
7281519	12/2006	Schroeder	N/A	N/A
7388303	12/2007	Seiver	N/A	N/A
7404294	12/2007	Sundin	N/A	N/A
7442239	12/2007	Armstrong et al.	N/A	N/A
7524173	12/2008	Cummins	N/A	N/A
7545130	12/2008	Latham	N/A	N/A
7552903	12/2008	Dunn et al.	N/A	N/A
7563076	12/2008	Brunet et al.	N/A	N/A

7563413	12/2008	Naets et al.	N/A	N/A
7574325	12/2008	Dykstra	N/A	N/A
7581379	12/2008	Yoshida et al.	N/A	N/A
7594424	12/2008	Fazekas	N/A	N/A
7614239	12/2008	Herzog et al.	N/A	N/A
7627416	12/2008	Batenburg et al.	N/A	N/A
7677316	12/2009	Butler et al.	N/A	N/A
7721521	12/2009	Kunkle et al.	N/A	N/A
7730711	12/2009	Kunkle et al.	N/A	N/A
7779961	12/2009	Matte	N/A	N/A
7789452	12/2009	Dempsey et al.	N/A	N/A
7836949	12/2009	Dykstra	N/A	N/A
7841394	12/2009	McNeel et al.	N/A	N/A
7845413	12/2009	Shampine et al.	N/A	N/A
7861679	12/2010	Lemke et al.	N/A	N/A
7886702	12/2010	Jerrell et al.	N/A	N/A
7900724	12/2010	Promersberger et al.	N/A	N/A
7921914	12/2010	Bruins et al.	N/A	N/A
7938151	12/2010	Höckner	N/A	N/A
7955056	12/2010	Pettersson	N/A	N/A
7980357	12/2010	Edwards	N/A	N/A
8056635	12/2010	Shampine et al.	N/A	N/A
8083504	12/2010	Williams et al.	N/A	N/A
8099942	12/2011	Alexander	N/A	N/A
8186334	12/2011	Ooyama	N/A	N/A
8196555	12/2011	Ikeda et al.	N/A	N/A
8202354	12/2011	Iijima	N/A	N/A
8316936	12/2011	Roddy et al.	N/A	N/A
8336631	12/2011	Shampine et al.	N/A	N/A
8388317	12/2012	Sung	N/A	N/A
8414673	12/2012	Raje et al.	N/A	N/A
8469826	12/2012	Brosowske	N/A	N/A
8500215	12/2012	Gastauer	N/A	N/A
8506267	12/2012	Gambier et al.	N/A	N/A
8575873	12/2012	Peterson et al.	N/A	N/A
8616005	12/2012	Cousino, Sr. et al.	N/A	N/A
8621873	12/2013	Robertson et al.	N/A	N/A
8641399	12/2013	Mucibabic	N/A	N/A
8656990	12/2013	Kajaria et al.	N/A	N/A
8672606	12/2013	Glynn et al.	N/A	N/A
8707853	12/2013	Dille et al.	N/A	N/A
8708667	12/2013	Collingborn	N/A	N/A
8714253	12/2013	Sherwood et al.	N/A	N/A
8757918	12/2013	Ramnarain et al.	N/A	N/A
8763583	12/2013	Hofbauer et al.	N/A	N/A
8770329	12/2013	Spitler	N/A	N/A
8784081	12/2013	Blume	N/A	N/A
8789601	12/2013	Broussard et al.	N/A	N/A
8794307	12/2013	Coquilleau et al.	N/A	N/A
8801394	12/2013	Anderson	N/A	N/A
8851186	12/2013	Shampine et al.	N/A	N/A
8851441	12/2013	Acuna et al.	N/A	N/A
8894356	12/2013	Lafontaine et al.	N/A	N/A
8905056	12/2013	Kendrick	N/A	N/A
8951019	12/2014	Hains et al.	N/A	N/A
8973560	12/2014	Krug	N/A	N/A
8997904	12/2014	Cryer et al.	N/A	N/A
9011111	12/2014	Lesko	N/A	N/A
9016383	12/2014	Shampine et al.	N/A	N/A
9032620	12/2014	Frassinelli et al.	N/A	N/A
9057247	12/2014	Kumar et al.	N/A	N/A
9097249	12/2014	Petersen	N/A	N/A
9103193	12/2014	Coli et al.	N/A	N/A
9121257	12/2014	Coli et al.	N/A	N/A
9140110	12/2014	Coli et al.	N/A	N/A
9175810	12/2014	Hains	N/A	N/A
9187982	12/2014	Dehring et al.	N/A	N/A
9206667	12/2014	Khvoshchev et al.	N/A	N/A
9212643	12/2014	Deliyski	N/A	N/A
9222346	12/2014	Walls	N/A	N/A
9324049	12/2015	Thomeer et al.	N/A	N/A

9341055	12/2015	Weightman et al.	N/A	N/A
9346662	12/2015	Van Vliet et al.	N/A	N/A
9366114	12/2015	Coli et al.	N/A	N/A
9376786	12/2015	Numasawa	N/A	N/A
9394829	12/2015	Cabeen et al.	N/A	N/A
9395049	12/2015	Vicknair et al.	N/A	N/A
9401670	12/2015	Minato et al.	N/A	N/A
9410410	12/2015	Broussard et al.	N/A	N/A
9410546	12/2015	Jaeger et al.	N/A	N/A
9429078	12/2015	Crowe et al.	N/A	N/A
9435333	12/2015	McCoy et al.	N/A	N/A
9488169	12/2015	Cochran et al.	N/A	N/A
9493997	12/2015	Liu et al.	N/A	N/A
9512783	12/2015	Veilleux et al.	N/A	N/A
9534473	12/2016	Morris et al.	N/A	N/A
9546652	12/2016	Yin	N/A	N/A
9550501	12/2016	Ledbetter	N/A	N/A
9556721	12/2016	Jang et al.	N/A	N/A
9562420	12/2016	Morris et al.	N/A	N/A
9570945	12/2016	Fischer	N/A	N/A
9579980	12/2016	Cryer et al.	N/A	N/A
9587649	12/2016	Oehring	N/A	N/A
9593710	12/2016	Laimboeck et al.	N/A	N/A
9611728	12/2016	Oehring	N/A	N/A
9617808	12/2016	Liu et al.	N/A	N/A
9638101	12/2016	Crowe et al.	N/A	N/A
9638194	12/2016	Wiegman et al.	N/A	N/A
9650871	12/2016	Oehring et al.	N/A	N/A
9656762	12/2016	Kamath et al.	N/A	N/A
9689316	12/2016	Crom	N/A	N/A
9695808	12/2016	Giessbach et al.	N/A	N/A
9739130	12/2016	Young	N/A	N/A
9764266	12/2016	Carter	N/A	N/A
9777748	12/2016	Lu et al.	N/A	N/A
9803467	12/2016	Tang et al.	N/A	N/A
9803793	12/2016	Davi et al.	N/A	N/A
9809308	12/2016	Aguilar et al.	N/A	N/A
9829002	12/2016	Crom	N/A	N/A
9840897	12/2016	Larson	N/A	N/A
9840901	12/2016	Oering et al.	N/A	N/A
9845730	12/2016	Betti et al.	N/A	N/A
9850422	12/2016	Lestz et al.	N/A	N/A
9856131	12/2017	Moffitt	N/A	N/A
9863279	12/2017	Laing et al.	N/A	N/A
9869305	12/2017	Crowe et al.	N/A	N/A
9871406	12/2017	Churnock et al.	N/A	N/A
9879609	12/2017	Crowe et al.	N/A	N/A
RE46725	12/2017	Case et al.	N/A	N/A
9893500	12/2017	Oehring et al.	N/A	N/A
9893660	12/2017	Peterson et al.	N/A	N/A
9897003	12/2017	Motakef et al.	N/A	N/A
9920615	12/2017	Zhang et al.	N/A	N/A
9945365	12/2017	Hernandez et al.	N/A	N/A
9964052	12/2017	Millican et al.	N/A	N/A
9970278	12/2017	Broussard et al.	N/A	N/A
9981840	12/2017	Shock	N/A	N/A
9995102	12/2017	Dillie et al.	N/A	N/A
9995218	12/2017	Oehring et al.	N/A	N/A
10008880	12/2017	Vicknair et al.	N/A	N/A
10008912	12/2017	Davey et al.	N/A	N/A
10018096	12/2017	Wallimann et al.	N/A	N/A
10020711	12/2017	Oehring et al.	N/A	N/A
10024123	12/2017	Steffenhagen et al.	N/A	N/A
10029289	12/2017	Wendorski et al.	N/A	N/A
10030579	12/2017	Austin et al.	N/A	N/A
10036238	12/2017	Oehring	N/A	N/A
10040541	12/2017	Wilson et al.	N/A	N/A
10060293	12/2017	Del Bono	N/A	N/A
10060349	12/2017	Álvarez et al.	N/A	N/A
10077933	12/2017	Nelson et al.	N/A	N/A
10082137	12/2017	Graham et al.	N/A	N/A

10094366	12/2017	Marica	N/A	N/A
10100827	12/2017	Devan et al.	N/A	N/A
10107084	12/2017	Coli et al.	N/A	N/A
10107085	12/2017	Coli et al.	N/A	N/A
10114061	12/2017	Frampton et al.	N/A	N/A
10119381	12/2017	Oehring et al.	N/A	N/A
10125750	12/2017	Pfaff	N/A	N/A
10134257	12/2017	Zhang et al.	N/A	N/A
10138098	12/2017	Sorensen et al.	N/A	N/A
10151244	12/2017	Giancotti et al.	N/A	N/A
10161423	12/2017	Rampen	N/A	N/A
10174599	12/2018	Shampine et al.	N/A	N/A
10184397	12/2018	Austin et al.	N/A	N/A
10196258	12/2018	Kalala et al.	N/A	N/A
10221856	12/2018	Hernandez et al.	N/A	N/A
10227854	12/2018	Glass	N/A	N/A
10227855	12/2018	Coli et al.	N/A	N/A
10246984	12/2018	Payne et al.	N/A	N/A
10247182	12/2018	Zhang et al.	N/A	N/A
10254732	12/2018	Oehring et al.	N/A	N/A
10267439	12/2018	Pryce et al.	N/A	N/A
10280724	12/2018	Hinderliter	N/A	N/A
10287943	12/2018	Schiltz	N/A	N/A
10288519	12/2018	De La Cruz	N/A	N/A
10303190	12/2018	Shock	N/A	N/A
10305350	12/2018	Johnson et al.	N/A	N/A
10316832	12/2018	Byrne	N/A	N/A
10317875	12/2018	Pandurangan et al.	N/A	N/A
10337402	12/2018	Austin et al.	N/A	N/A
10358035	12/2018	Cryer	N/A	N/A
10371012	12/2018	Davis et al.	N/A	N/A
10374485	12/2018	Morris et al.	N/A	N/A
10378326	12/2018	Morris et al.	N/A	N/A
10393108	12/2018	Chong et al.	N/A	N/A
10407990	12/2018	Oehring et al.	N/A	N/A
10408031	12/2018	Oehring et al.	N/A	N/A
10415348	12/2018	Zhang et al.	N/A	N/A
10415557	12/2018	Crowe et al.	N/A	N/A
10415562	12/2018	Kajita et al.	N/A	N/A
RE47695	12/2018	Case et al.	N/A	N/A
10465689	12/2018	Crom	N/A	N/A
10478753	12/2018	Elms et al.	N/A	N/A
10526882	12/2019	Oehring et al.	N/A	N/A
10563649	12/2019	Zhang et al.	N/A	N/A
10577910	12/2019	Stephenson	N/A	N/A
10584645	12/2019	Nakagawa et al.	N/A	N/A
10590867	12/2019	Thomassin et al.	N/A	N/A
10598258	12/2019	Oehring et al.	N/A	N/A
10610842	12/2019	Chong	N/A	N/A
10662749	12/2019	Hill et al.	N/A	N/A
10711787	12/2019	Darley	N/A	N/A
10738580	12/2019	Fischer et al.	N/A	N/A
10753153	12/2019	Fischer et al.	N/A	N/A
10753165	12/2019	Fischer et al.	N/A	N/A
10760556	12/2019	Crom et al.	N/A	N/A
10794165	12/2019	Fischer et al.	N/A	N/A
10794166	12/2019	Reckels et al.	N/A	N/A
10801311	12/2019	Cui et al.	N/A	N/A
10815764	12/2019	Yeung et al.	N/A	N/A
10815978	12/2019	Glass	N/A	N/A
10830032	12/2019	Zhang et al.	N/A	N/A
10830225	12/2019	Repaci	N/A	N/A
10859203	12/2019	Cui et al.	N/A	N/A
10864487	12/2019	Han et al.	N/A	N/A
10865624	12/2019	Cui et al.	N/A	N/A
10865631	12/2019	Zhang et al.	N/A	N/A
10870093	12/2019	Zhong et al.	N/A	N/A
10871045	12/2019	Fischer et al.	N/A	N/A
10892596	12/2020	Enya et al.	N/A	N/A
10895202	12/2020	Yeung et al.	N/A	N/A
10900475	12/2020	Weightman et al.	N/A	N/A

10907459	12/2020	Yeung et al.	N/A	N/A
10927774	12/2020	Cai et al.	N/A	N/A
10927802	12/2020	Oehring	N/A	N/A
10954770	12/2020	Yeung et al.	N/A	N/A
10954855	12/2020	Ji et al.	N/A	N/A
10961614	12/2020	Yeung et al.	N/A	N/A
10961908	12/2020	Yeung et al.	N/A	N/A
10961912	12/2020	Yeung et al.	N/A	N/A
10961914	12/2020	Yeung et al.	N/A	N/A
10961993	12/2020	Ji et al.	N/A	N/A
10961995	12/2020	Mayorca	N/A	N/A
10968837	12/2020	Yeung et al.	N/A	N/A
10982523	12/2020	Hill et al.	N/A	N/A
10989019	12/2020	Cai et al.	N/A	N/A
10989180	12/2020	Yeung et al.	N/A	N/A
10995564	12/2020	Miller et al.	N/A	N/A
11002189	12/2020	Yeung et al.	N/A	N/A
11008950	12/2020	Ethier et al.	N/A	N/A
11015423	12/2020	Yeung et al.	N/A	N/A
11015536	12/2020	Yeung et al.	N/A	N/A
11015594	12/2020	Yeung et al.	N/A	N/A
11022526	12/2020	Yeung et al.	N/A	N/A
11028677	12/2020	Yeung et al.	N/A	N/A
11035213	12/2020	Dusterhoft et al.	N/A	N/A
11035214	12/2020	Cui et al.	N/A	N/A
11047379	12/2020	Li et al.	N/A	N/A
11053853	12/2020	Li et al.	N/A	N/A
11060455	12/2020	Yeung et al.	N/A	N/A
11066915	12/2020	Yeung et al.	N/A	N/A
11068455	12/2020	Shabi et al.	N/A	N/A
11085281	12/2020	Yeung et al.	N/A	N/A
11085282	12/2020	Mazrooe et al.	N/A	N/A
11092152	12/2020	Yeung et al.	N/A	N/A
11098651	12/2020	Yeung et al.	N/A	N/A
11105250	12/2020	Zhang et al.	N/A	N/A
11105266	12/2020	Zhou et al.	N/A	N/A
11109508	12/2020	Yeung et al.	N/A	N/A
11111768	12/2020	Yeung et al.	N/A	N/A
11125066	12/2020	Yeung et al.	N/A	N/A
11125156	12/2020	Zhang et al.	N/A	N/A
11129295	12/2020	Yeung et al.	N/A	N/A
11143000	12/2020	Li et al.	N/A	N/A
11143005	12/2020	Dusterhoft et al.	N/A	N/A
11143006	12/2020	Zhang et al.	N/A	N/A
11149533	12/2020	Yeung et al.	N/A	N/A
11149726	12/2020	Yeung et al.	N/A	N/A
11156159	12/2020	Yeung et al.	N/A	N/A
11168681	12/2020	Boguski	N/A	N/A
11174716	12/2020	Yeung et al.	N/A	N/A
11193360	12/2020	Yeung et al.	N/A	N/A
11193361	12/2020	Yeung et al.	N/A	N/A
11205880	12/2020	Yeung et al.	N/A	N/A
11205881	12/2020	Yeung et al.	N/A	N/A
11208879	12/2020	Yeung et al.	N/A	N/A
11208953	12/2020	Yeung et al.	N/A	N/A
11220895	12/2021	Yeung et al.	N/A	N/A
11236739	12/2021	Yeung et al.	N/A	N/A
11242737	12/2021	Zhang et al.	N/A	N/A
11243509	12/2021	Cai et al.	N/A	N/A
11251650	12/2021	Liu et al.	N/A	N/A
11261717	12/2021	Yeung et al.	N/A	N/A
11268346	12/2021	Yeung et al.	N/A	N/A
11280266	12/2021	Yeung et al.	N/A	N/A
11306835	12/2021	Dille et al.	N/A	N/A
RE49083	12/2021	Case et al.	N/A	N/A
11339638	12/2021	Yeung et al.	N/A	N/A
11346200	12/2021	Cai et al.	N/A	N/A
11373058	12/2021	Jaaskelainen et al.	N/A	N/A
RE49140	12/2021	Case et al.	N/A	N/A
11377943	12/2021	Kriebel et al.	N/A	N/A
RE49155	12/2021	Case et al.	N/A	N/A

RE49156	12/2021	Case et al.	N/A	N/A
11401927	12/2021	Li et al.	N/A	N/A
11428165	12/2021	Yeung et al.	N/A	N/A
11441483	12/2021	Li et al.	N/A	N/A
11448122	12/2021	Feng et al.	N/A	N/A
11466680	12/2021	Yeung et al.	N/A	N/A
11480040	12/2021	Han et al.	N/A	N/A
11492887	12/2021	Cui et al.	N/A	N/A
11499405	12/2021	Zhang et al.	N/A	N/A
11506039	12/2021	Zhang et al.	N/A	N/A
11512570	12/2021	Yeung	N/A	N/A
11519395	12/2021	Zhang et al.	N/A	N/A
11519405	12/2021	Deng et al.	N/A	N/A
11530602	12/2021	Yeung et al.	N/A	N/A
11549349	12/2022	Wang et al.	N/A	N/A
11555390	12/2022	Cui et al.	N/A	N/A
11555756	12/2022	Yeung et al.	N/A	N/A
11557887	12/2022	Ji et al.	N/A	N/A
11560779	12/2022	Mao et al.	N/A	N/A
11560845	12/2022	Yeung et al.	N/A	N/A
11572775	12/2022	Mao et al.	N/A	N/A
11575249	12/2022	Ji et al.	N/A	N/A
11592020	12/2022	Chang et al.	N/A	N/A
11596047	12/2022	Liu et al.	N/A	N/A
11598263	12/2022	Yeung et al.	N/A	N/A
11603797	12/2022	Zhang et al.	N/A	N/A
11607982	12/2022	Tian et al.	N/A	N/A
11608726	12/2022	Zhang et al.	N/A	N/A
11624326	12/2022	Yeung et al.	N/A	N/A
11629583	12/2022	Yeung et al.	N/A	N/A
11629589	12/2022	Lin et al.	N/A	N/A
11649766	12/2022	Yeung et al.	N/A	N/A
11662384	12/2022	Liu et al.	N/A	N/A
11668173	12/2022	Zhang et al.	N/A	N/A
11668289	12/2022	Chang et al.	N/A	N/A
11677238	12/2022	Liu et al.	N/A	N/A
2002/0126922	12/2001	Cheng et al.	N/A	N/A
2002/0197176	12/2001	Kondo	N/A	N/A
2003/0031568	12/2002	Stiefel	N/A	N/A
2003/0061819	12/2002	Kuroki et al.	N/A	N/A
2003/0161212	12/2002	Neal et al.	N/A	N/A
2003/0218850	12/2002	Kubo	361/103	H01L 23/467
2004/0016245	12/2003	Pierson	N/A	N/A
2004/0074238	12/2003	Watanabe et al.	N/A	N/A
2004/0076526	12/2003	Fukano et al.	N/A	N/A
2004/0187950	12/2003	Cohen et al.	N/A	N/A
2004/0219040	12/2003	Kugelev et al.	N/A	N/A
2005/0051322	12/2004	Speer	N/A	N/A
2005/0056081	12/2004	Gocho	N/A	N/A
2005/0139286	12/2004	Poulter	N/A	N/A
2005/0196298	12/2004	Manning	N/A	N/A
2005/0226754	12/2004	Orr et al.	N/A	N/A
2005/0274134	12/2004	Ryu et al.	N/A	N/A
2006/0061091	12/2005	Osterloh	N/A	N/A
2006/0062914	12/2005	Garg et al.	N/A	N/A
2006/0196251	12/2005	Richey	N/A	N/A
2006/0211356	12/2005	Grassman	N/A	N/A
2006/0228225	12/2005	Rogers	N/A	N/A
2006/0260331	12/2005	Andreychuk	N/A	N/A
2006/0272333	12/2005	Sundin	N/A	N/A
2007/0029090	12/2006	Andreychuk et al.	N/A	N/A
2007/0041848	12/2006	Wood et al.	N/A	N/A
2007/0066406	12/2006	Keller et al.	N/A	N/A
2007/0098580	12/2006	Petersen	N/A	N/A
2007/0107981	12/2006	Sicotte	N/A	N/A
2007/0125544	12/2006	Robinson et al.	N/A	N/A
2007/0169543	12/2006	Fazekas	N/A	N/A
2007/0181212	12/2006	Fell	N/A	N/A
2007/0277982	12/2006	Shampine et al.	N/A	N/A
2007/0295569	12/2006	Manzoor et al.	N/A	N/A
2008/0006089	12/2007	Adnan et al.	N/A	N/A

2008/0098891	12/2007	Feher	N/A	N/A
2008/0161974	12/2007	Alston	N/A	N/A
2008/0212275	12/2007	Waryck et al.	N/A	N/A
2008/0229757	12/2007	Alexander et al.	N/A	N/A
2008/0264625	12/2007	Ochoa	N/A	N/A
2008/0264649	12/2007	Crawford	N/A	N/A
2008/0298982	12/2007	Pabst	N/A	N/A
2009/0064685	12/2008	Busekros et al.	N/A	N/A
2009/0068031	12/2008	Gambier et al.	N/A	N/A
2009/0092510	12/2008	Williams et al.	N/A	N/A
2009/0124191	12/2008	Van Becelaere et al.	N/A	N/A
2009/0178412	12/2008	Spytek	N/A	N/A
2009/0212630	12/2008	Flegel et al.	N/A	N/A
2009/0249794	12/2008	Wilkes et al.	N/A	N/A
2009/0252616	12/2008	Brunet et al.	N/A	N/A
2009/0308602	12/2008	Bruins et al.	N/A	N/A
2010/0019626	12/2009	Stout et al.	N/A	N/A
2010/0071899	12/2009	Coquilleau et al.	N/A	N/A
2010/0218508	12/2009	Brown et al.	N/A	N/A
2010/0300683	12/2009	Looper et al.	N/A	N/A
2010/0310384	12/2009	Stephenson et al.	N/A	N/A
2011/0041681	12/2010	Duerr	N/A	N/A
2011/0052423	12/2010	Gambier et al.	N/A	N/A
2011/0054704	12/2010	Karpman et al.	N/A	N/A
2011/0085924	12/2010	Shampine et al.	N/A	N/A
2011/0146244	12/2010	Farman et al.	N/A	N/A
2011/0146246	12/2010	Farman et al.	N/A	N/A
2011/0173991	12/2010	Dean	N/A	N/A
2011/0197988	12/2010	Van Vliet et al.	N/A	N/A
2011/0241888	12/2010	Lu et al.	N/A	N/A
2011/0265443	12/2010	Ansari	N/A	N/A
2011/0272158	12/2010	Neal	N/A	N/A
2012/0023973	12/2011	Mayorca	N/A	N/A
2012/0048242	12/2011	Sumilla et al.	N/A	N/A
2012/0085541	12/2011	Love et al.	N/A	N/A
2012/0137699	12/2011	Montagne et al.	N/A	N/A
2012/0179444	12/2011	Ganguly et al.	N/A	N/A
2012/0192542	12/2011	Chillar et al.	N/A	N/A
2012/0199001	12/2011	Chillar et al.	N/A	N/A
2012/0204627	12/2011	Anderl et al.	N/A	N/A
2012/0255734	12/2011	Coli et al.	N/A	N/A
2012/0310509	12/2011	Pardo et al.	N/A	N/A
2012/0324903	12/2011	Dewis et al.	N/A	N/A
2013/0068307	12/2012	Hains et al.	N/A	N/A
2013/0087045	12/2012	Sullivan et al.	N/A	N/A
2013/0087945	12/2012	Kusters et al.	N/A	N/A
2013/0134702	12/2012	Boraas et al.	N/A	N/A
2013/0189915	12/2012	Hazard	N/A	N/A
2013/0205798	12/2012	Kwok et al.	N/A	N/A
2013/0233165	12/2012	Matzner et al.	N/A	N/A
2013/0255953	12/2012	Tudor	N/A	N/A
2013/0259707	12/2012	Yin	N/A	N/A
2013/0284455	12/2012	Kajaria et al.	N/A	N/A
2013/0300341	12/2012	Gillette	N/A	N/A
2013/0306322	12/2012	Sanborn	N/A	N/A
2014/0000668	12/2013	Lessard	N/A	N/A
2014/0010671	12/2013	Cryer et al.	N/A	N/A
2014/0013768	12/2013	Laing et al.	N/A	N/A
2014/0032082	12/2013	Gehrke et al.	N/A	N/A
2014/0044517	12/2013	Saha et al.	N/A	N/A
2014/0048253	12/2013	Andreychuk	N/A	N/A
2014/0090729	12/2013	Coulter et al.	N/A	N/A
2014/0090742	12/2013	Coskrey et al.	N/A	N/A
2014/0094105	12/2013	Lundh et al.	N/A	N/A
2014/0095114	12/2013	Thomeer et al.	N/A	N/A
2014/0095554	12/2013	Thomeer et al.	N/A	N/A
2014/0123621	12/2013	Driessens et al.	N/A	N/A
2014/0130422	12/2013	Laing et al.	N/A	N/A
2014/0138079	12/2013	Broussard et al.	N/A	N/A
2014/0144641	12/2013	Chandler	N/A	N/A
2014/0147291	12/2013	Burnette	N/A	N/A

2014/0158345	12/2013	Jang et al.	N/A	N/A
2014/0174097	12/2013	Hammer et al.	N/A	N/A
2014/0196459	12/2013	Futa et al.	N/A	N/A
2014/0216736	12/2013	Leugemors et al.	N/A	N/A
2014/0219824	12/2013	Burnette	N/A	N/A
2014/0250845	12/2013	Jackson et al.	N/A	N/A
2014/0251623	12/2013	Lestz et al.	N/A	N/A
2014/0277772	12/2013	Lopez et al.	N/A	N/A
2014/0290266	12/2013	Veilleux, Jr. et al.	N/A	N/A
2014/0318638	12/2013	Harwood et al.	N/A	N/A
2014/0322050	12/2013	Marette et al.	N/A	N/A
2015/0027730	12/2014	Hall et al.	N/A	N/A
2015/0078924	12/2014	Zhang et al.	N/A	N/A
2015/0101344	12/2014	Jarrier et al.	N/A	N/A
2015/0114652	12/2014	Lestz et al.	N/A	N/A
2015/0129210	12/2014	Chong et al.	N/A	N/A
2015/0135659	12/2014	Jarrier et al.	N/A	N/A
2015/0159553	12/2014	Kippel et al.	N/A	N/A
2015/0192117	12/2014	Bridges	N/A	N/A
2015/0204148	12/2014	Liu et al.	N/A	N/A
2015/0204322	12/2014	Iund et al.	N/A	N/A
2015/0211512	12/2014	Wiegman et al.	N/A	N/A
2015/0214816	12/2014	Raad	N/A	N/A
2015/0217672	12/2014	Shampine et al.	N/A	N/A
2015/0226140	12/2014	Zhang et al.	N/A	N/A
2015/0252661	12/2014	Glass	N/A	N/A
2015/0275891	12/2014	Chong et al.	N/A	N/A
2015/0337730	12/2014	Kupiszewski et al.	N/A	N/A
2015/0340864	12/2014	Compton	N/A	N/A
2015/0345385	12/2014	Santini	N/A	N/A
2015/0369351	12/2014	Hermann et al.	N/A	N/A
2016/0032703	12/2015	Broussard et al.	N/A	N/A
2016/0032836	12/2015	Hawkinson et al.	N/A	N/A
2016/0076447	12/2015	Merlo et al.	N/A	N/A
2016/0102581	12/2015	Del Bono	N/A	N/A
2016/0105022	12/2015	Oehring et al.	N/A	N/A
2016/0108713	12/2015	Dunaeva et al.	N/A	N/A
2016/0123185	12/2015	Le Pache et al.	N/A	N/A
2016/0168979	12/2015	Zhang et al.	N/A	N/A
2016/0177675	12/2015	Morris et al.	N/A	N/A
2016/0177945	12/2015	Byrne et al.	N/A	N/A
2016/0186671	12/2015	Austin et al.	N/A	N/A
2016/0195082	12/2015	Wiegman et al.	N/A	N/A
2016/0215774	12/2015	Oklejas et al.	N/A	N/A
2016/0230525	12/2015	Lestz et al.	N/A	N/A
2016/0244314	12/2015	Van Vliet et al.	N/A	N/A
2016/0248230	12/2015	Tawy et al.	N/A	N/A
2016/0253634	12/2015	Thomeer et al.	N/A	N/A
2016/0258267	12/2015	Payne et al.	N/A	N/A
2016/0273328	12/2015	Oehring	N/A	N/A
2016/0273346	12/2015	Tang et al.	N/A	N/A
2016/0290114	12/2015	Oehring et al.	N/A	N/A
2016/0319650	12/2015	Oehring et al.	N/A	N/A
2016/0326845	12/2015	Djikpesse et al.	N/A	N/A
2016/0348479	12/2015	Oehring et al.	N/A	N/A
2016/0369609	12/2015	Morris et al.	N/A	N/A
2017/0009905	12/2016	Arnold	N/A	N/A
2017/0016433	12/2016	Chong et al.	N/A	N/A
2017/0030177	12/2016	Oehring et al.	N/A	N/A
2017/0038137	12/2016	Turney	N/A	N/A
2017/0045055	12/2016	Hoefel et al.	N/A	N/A
2017/0052087	12/2016	Faqihi et al.	N/A	N/A
2017/0074074	12/2016	Joseph et al.	N/A	N/A
2017/0074076	12/2016	Joseph et al.	N/A	N/A
2017/0074089	12/2016	Agarwal et al.	N/A	N/A
2017/0082110	12/2016	Lammers	N/A	N/A
2017/0089189	12/2016	Norris et al.	N/A	N/A
2017/0114613	12/2016	Lecerf et al.	N/A	N/A
2017/0114625	12/2016	Norris et al.	N/A	N/A
2017/0122310	12/2016	Ladron de Guevara	N/A	N/A
2017/0131174	12/2016	Enev et al.	N/A	N/A

2017/0145918	12/2016	Oehring et al.	N/A	N/A
2017/0191350	12/2016	Johns et al.	N/A	N/A
2017/0218727	12/2016	Oehring et al.	N/A	N/A
2017/0226839	12/2016	Broussard et al.	N/A	N/A
2017/0226842	12/2016	Omont et al.	N/A	N/A
2017/0226998	12/2016	Zhang et al.	N/A	N/A
2017/0227002	12/2016	Mikulski et al.	N/A	N/A
2017/0233103	12/2016	Teicholz et al.	N/A	N/A
2017/0234165	12/2016	Kersey	60/798	F02C 7/20
2017/0234308	12/2016	Buckley	N/A	N/A
2017/0241336	12/2016	Jones et al.	N/A	N/A
2017/0241671	12/2016	Ahmad	N/A	N/A
2017/0248034	12/2016	Dzieciol et al.	N/A	N/A
2017/0248208	12/2016	Tamura	N/A	N/A
2017/0248308	12/2016	Makarychev-Mikhailov et al.	N/A	N/A
2017/0275149	12/2016	Schmidt	N/A	N/A
2017/0288400	12/2016	Williams	N/A	N/A
2017/0292409	12/2016	Aguilar et al.	N/A	N/A
2017/0302135	12/2016	Cory	N/A	N/A
2017/0305736	12/2016	Haile et al.	N/A	N/A
2017/0306847	12/2016	Suciu et al.	N/A	N/A
2017/0306936	12/2016	Dole	N/A	N/A
2017/0322086	12/2016	Luharuka	N/A	N/A
2017/0333086	12/2016	Jackson	N/A	N/A
2017/0334448	12/2016	Schwunk	N/A	N/A
2017/0335842	12/2016	Robinson et al.	N/A	N/A
2017/0350471	12/2016	Steidl et al.	N/A	N/A
2017/0356470	12/2016	Jaffrey	N/A	N/A
2017/0370199	12/2016	Witkowski et al.	N/A	N/A
2017/0370480	12/2016	Witkowski et al.	N/A	N/A
2018/0034280	12/2017	Pedersen	N/A	N/A
2018/0038216	12/2017	Zhang et al.	N/A	N/A
2018/0038328	12/2017	Louven et al.	N/A	N/A
2018/0041093	12/2017	Miranda	N/A	N/A
2018/0045202	12/2017	Crom	N/A	N/A
2018/0058171	12/2017	Roesner et al.	N/A	N/A
2018/0087499	12/2017	Zhang et al.	N/A	N/A
2018/0087996	12/2017	De La Cruz	N/A	N/A
2018/0156210	12/2017	Oehring et al.	N/A	N/A
2018/0172294	12/2017	Owen	N/A	N/A
2018/0183219	12/2017	Oehring et al.	N/A	N/A
2018/0186442	12/2017	Maier	N/A	N/A
2018/0187662	12/2017	Hill et al.	N/A	N/A
2018/0209415	12/2017	Zhang et al.	N/A	N/A
2018/0223640	12/2017	Keihany et al.	N/A	N/A
2018/0224044	12/2017	Penney	N/A	N/A
2018/0229998	12/2017	Shock	N/A	N/A
2018/0258746	12/2017	Broussard et al.	N/A	N/A
2018/0266412	12/2017	Stokkevag et al.	N/A	N/A
2018/0278124	12/2017	Oehring et al.	N/A	N/A
2018/0283102	12/2017	Cook	N/A	N/A
2018/0283618	12/2017	Cook	N/A	N/A
2018/0284817	12/2017	Cook et al.	N/A	N/A
2018/0290877	12/2017	Shock	N/A	N/A
2018/0291781	12/2017	Pedrini	N/A	N/A
2018/0298731	12/2017	Bishop	N/A	N/A
2018/0298735	12/2017	Conrad	N/A	N/A
2018/0307255	12/2017	Bishop	N/A	N/A
2018/0313456	12/2017	Bayyouk et al.	N/A	N/A
2018/0328157	12/2017	Bishop	N/A	F04B 17/06
2018/0334893	12/2017	Oehring	N/A	N/A
2018/0363435	12/2017	Coli et al.	N/A	N/A
2018/0363436	12/2017	Coli et al.	N/A	N/A
2018/0363437	12/2017	Coli et al.	N/A	N/A
2018/0363438	12/2017	Coli et al.	N/A	N/A
2019/0003272	12/2018	Morris et al.	N/A	N/A
2019/0003329	12/2018	Morris et al.	N/A	N/A
2019/0009232	12/2018	Yeung	N/A	B01F 35/2203
2019/0010793	12/2018	Hinderliter	N/A	N/A
2019/0011051	12/2018	Yeung	N/A	N/A
2019/0048993	12/2018	Akiyama et al.	N/A	N/A

2019/0063263	12/2018	Davis et al.	N/A	N/A
2019/0063341	12/2018	Davis	N/A	N/A
2019/0067991	12/2018	Davis et al.	N/A	N/A
2019/0071992	12/2018	Feng	N/A	N/A
2019/0072005	12/2018	Fisher et al.	N/A	N/A
2019/0078471	12/2018	Braglia et al.	N/A	N/A
2019/0088845	12/2018	Sugi et al.	N/A	N/A
2019/0091619	12/2018	Huang	N/A	N/A
2019/0106316	12/2018	Van Vliet et al.	N/A	N/A
2019/0106970	12/2018	Oehring	N/A	N/A
2019/0112908	12/2018	Coli et al.	N/A	N/A
2019/0112910	12/2018	Oehring et al.	N/A	N/A
2019/0119096	12/2018	Haile et al.	N/A	N/A
2019/0120024	12/2018	Oehring et al.	N/A	N/A
2019/0120031	12/2018	Gilje	N/A	N/A
2019/0120134	12/2018	Goleczka et al.	N/A	N/A
2019/0128247	12/2018	Douglas, III	N/A	N/A
2019/0128288	12/2018	Konada et al.	N/A	N/A
2019/0131607	12/2018	Gillette	N/A	N/A
2019/0136677	12/2018	Shampine et al.	N/A	N/A
2019/0153843	12/2018	Headrick	N/A	N/A
2019/0153938	12/2018	Hammoud	N/A	N/A
2019/0154020	12/2018	Glass	N/A	N/A
2019/0155318	12/2018	Meunier	N/A	N/A
2019/0178234	12/2018	Beisel	N/A	N/A
2019/0178235	12/2018	Coskrey et al.	N/A	N/A
2019/0185312	12/2018	Bush et al.	N/A	N/A
2019/0203572	12/2018	Morris et al.	N/A	N/A
2019/0204021	12/2018	Morris et al.	N/A	N/A
2019/0211661	12/2018	Reckles et al.	N/A	N/A
2019/0211814	12/2018	Weightman et al.	N/A	N/A
2019/0217258	12/2018	Bishop	N/A	N/A
2019/0226317	12/2018	Payne et al.	N/A	N/A
2019/0245348	12/2018	Hinderliter et al.	N/A	N/A
2019/0249652	12/2018	Stephenson et al.	N/A	N/A
2019/0249754	12/2018	Oehring et al.	N/A	N/A
2019/0257297	12/2018	Botting et al.	N/A	N/A
2019/0264667	12/2018	Byrne	N/A	N/A
2019/0277279	12/2018	Byrne et al.	N/A	N/A
2019/0277295	12/2018	Clyburn et al.	N/A	N/A
2019/0309585	12/2018	Miller et al.	N/A	N/A
2019/0316447	12/2018	Oehring et al.	N/A	N/A
2019/0316456	12/2018	Beisel et al.	N/A	N/A
2019/0323337	12/2018	Glass et al.	N/A	N/A
2019/0330923	12/2018	Gable et al.	N/A	N/A
2019/0331117	12/2018	Gable et al.	N/A	N/A
2019/0337392	12/2018	Joshi et al.	N/A	N/A
2019/0338762	12/2018	Curry et al.	N/A	N/A
2019/0345920	12/2018	Surjaatmadja et al.	N/A	N/A
2019/0353103	12/2018	Roberge	N/A	N/A
2019/0356199	12/2018	Morris et al.	N/A	N/A
2019/0376449	12/2018	Carrell	N/A	N/A
2019/0383123	12/2018	Hinderliter	N/A	N/A
2020/0003205	12/2019	Stokkevåg et al.	N/A	N/A
2020/0011165	12/2019	George et al.	N/A	N/A
2020/0025058	12/2019	Jochman	N/A	F01P 11/10
2020/0040878	12/2019	Morris	N/A	N/A
2020/0049136	12/2019	Stephenson	N/A	N/A
2020/0049153	12/2019	Headrick et al.	N/A	N/A
2020/0071998	12/2019	Oehring et al.	N/A	N/A
2020/0072201	12/2019	Marica	N/A	N/A
2020/0088202	12/2019	Sigmar et al.	N/A	N/A
2020/0095854	12/2019	Hinderliter	N/A	N/A
2020/0109610	12/2019	Husoy et al.	N/A	N/A
2020/0109616	12/2019	Oehring et al.	N/A	N/A
2020/0123729	12/2019	Guender	N/A	E02D 27/44
2020/0132058	12/2019	Mollatt	N/A	N/A
2020/0141219	12/2019	Oehring et al.	N/A	N/A
2020/0141326	12/2019	Redford et al.	N/A	N/A
2020/0141907	12/2019	Meck et al.	N/A	N/A
2020/0166026	12/2019	Marica	N/A	N/A

2020/0206704	12/2019	Chong	N/A	N/A
2020/0208733	12/2019	Kim	N/A	N/A
2020/0223648	12/2019	Herman et al.	N/A	N/A
2020/0224645	12/2019	Buckley	N/A	N/A
2020/0232454	12/2019	Chretien et al.	N/A	N/A
2020/0256333	12/2019	Surjaatmadja	N/A	N/A
2020/0263498	12/2019	Fischer et al.	N/A	N/A
2020/0263525	12/2019	Reid	N/A	N/A
2020/0263526	12/2019	Fischer et al.	N/A	N/A
2020/0263527	12/2019	Fischer et al.	N/A	N/A
2020/0263528	12/2019	Fischer et al.	N/A	N/A
2020/0267888	12/2019	Putz	N/A	N/A
2020/0291731	12/2019	Haiderer et al.	N/A	N/A
2020/0295574	12/2019	Batsch-Smith	N/A	N/A
2020/0300050	12/2019	Oehring et al.	N/A	N/A
2020/0309027	12/2019	Rytkonen	N/A	N/A
2020/0309113	12/2019	Hunter et al.	N/A	N/A
2020/0325752	12/2019	Clark et al.	N/A	N/A
2020/0325760	12/2019	Markham	N/A	N/A
2020/0325761	12/2019	Williams	N/A	N/A
2020/0325791	12/2019	Himmelmann	N/A	N/A
2020/0325893	12/2019	Kraïge et al.	N/A	N/A
2020/0332784	12/2019	Zhang et al.	N/A	N/A
2020/0332788	12/2019	Cui et al.	N/A	N/A
2020/0340313	12/2019	Fischer et al.	N/A	N/A
2020/0340322	12/2019	Sizemore et al.	N/A	N/A
2020/0340340	12/2019	Oehring et al.	N/A	N/A
2020/0340344	12/2019	Reckels et al.	N/A	N/A
2020/0340404	12/2019	Stockstill	N/A	N/A
2020/0347725	12/2019	Morris et al.	N/A	N/A
2020/0354928	12/2019	Wehler et al.	N/A	N/A
2020/0355055	12/2019	Dusterhoft et al.	N/A	N/A
2020/0362760	12/2019	Morenko et al.	N/A	N/A
2020/0362764	12/2019	Saintignan et al.	N/A	N/A
2020/0370394	12/2019	Cai et al.	N/A	N/A
2020/0370408	12/2019	Cai et al.	N/A	N/A
2020/0370429	12/2019	Cai et al.	N/A	N/A
2020/0371490	12/2019	Cai et al.	N/A	N/A
2020/0386169	12/2019	Hinderliter et al.	N/A	N/A
2020/0386222	12/2019	Pham et al.	N/A	N/A
2020/0388140	12/2019	Gomez et al.	N/A	N/A
2020/0392826	12/2019	Cui et al.	N/A	N/A
2020/0392827	12/2019	George et al.	N/A	N/A
2020/0393088	12/2019	Sizemore et al.	N/A	N/A
2020/0398238	12/2019	Zhong et al.	N/A	N/A
2020/0400000	12/2019	Ghasripoor et al.	N/A	N/A
2020/0400005	12/2019	Han et al.	N/A	N/A
2020/0407625	12/2019	Stephenson	N/A	N/A
2020/0408071	12/2019	Li et al.	N/A	N/A
2020/0408144	12/2019	Feng et al.	N/A	N/A
2020/0408147	12/2019	Zhang et al.	N/A	N/A
2020/0408149	12/2019	Li et al.	N/A	N/A
2021/0025324	12/2020	Morris et al.	N/A	N/A
2021/0025383	12/2020	Bodishbaugh et al.	N/A	N/A
2021/0032961	12/2020	Hinderliter et al.	N/A	N/A
2021/0054727	12/2020	Floyd	N/A	N/A
2021/0071503	12/2020	Ogg et al.	N/A	N/A
2021/0071574	12/2020	Feng et al.	N/A	N/A
2021/0071579	12/2020	Li et al.	N/A	N/A
2021/0071654	12/2020	Brunson	N/A	N/A
2021/0071752	12/2020	Cui et al.	N/A	N/A
2021/0079758	12/2020	Yeung et al.	N/A	N/A
2021/0079851	12/2020	Yeung et al.	N/A	N/A
2021/0086851	12/2020	Zhang et al.	N/A	N/A
2021/0087883	12/2020	Zhang et al.	N/A	N/A
2021/0087916	12/2020	Zhang et al.	N/A	N/A
2021/0087925	12/2020	Heidari et al.	N/A	N/A
2021/0087943	12/2020	Cui et al.	N/A	N/A
2021/0088042	12/2020	Zhang et al.	N/A	N/A
2021/0123425	12/2020	Cui et al.	N/A	N/A
2021/0123434	12/2020	Cui et al.	N/A	N/A

2021/0123435	12/2020	Cui et al.	N/A	N/A
2021/0131409	12/2020	Cui et al.	N/A	N/A
2021/0140416	12/2020	Buckley	N/A	N/A
2021/0148208	12/2020	Thomas et al.	N/A	N/A
2021/0156240	12/2020	Cicci et al.	N/A	N/A
2021/0156241	12/2020	Cook	N/A	N/A
2021/0172282	12/2020	Wang et al.	N/A	N/A
2021/0180517	12/2020	Zhou et al.	N/A	N/A
2021/0190045	12/2020	Zhang et al.	N/A	N/A
2021/0199110	12/2020	Albert et al.	N/A	N/A
2021/0222690	12/2020	Beisel	N/A	N/A
2021/0239112	12/2020	Buckley	N/A	N/A
2021/0246774	12/2020	Cui et al.	N/A	N/A
2021/0270261	12/2020	Zhang et al.	N/A	N/A
2021/0270264	12/2020	Byrne	N/A	N/A
2021/0285311	12/2020	Ji et al.	N/A	N/A
2021/0285432	12/2020	Ji et al.	N/A	N/A
2021/0301807	12/2020	Cui et al.	N/A	N/A
2021/0306720	12/2020	Sandoval et al.	N/A	N/A
2021/0308638	12/2020	Zhong et al.	N/A	N/A
2021/0324718	12/2020	Anders	N/A	N/A
2021/0348475	12/2020	Yeung et al.	N/A	N/A
2021/0348476	12/2020	Yeung et al.	N/A	N/A
2021/0348477	12/2020	Yeung et al.	N/A	N/A
2021/0355927	12/2020	Jian et al.	N/A	N/A
2021/0372394	12/2020	Bagulayan et al.	N/A	N/A
2021/0372395	12/2020	Li et al.	N/A	N/A
2021/0376413	12/2020	Asfha	N/A	N/A
2021/0388760	12/2020	Feng et al.	N/A	N/A
2022/0082007	12/2021	Zhang et al.	N/A	N/A
2022/0090476	12/2021	Zhang et al.	N/A	N/A
2022/0090477	12/2021	Zhang et al.	N/A	N/A
2022/0090478	12/2021	Zhang et al.	N/A	N/A
2022/0112892	12/2021	Cui et al.	N/A	N/A
2022/0120262	12/2021	Ji et al.	N/A	N/A
2022/0145740	12/2021	Yuan et al.	N/A	N/A
2022/0154775	12/2021	Liu et al.	N/A	N/A
2022/0155373	12/2021	Liu et al.	N/A	N/A
2022/0162931	12/2021	Zhong et al.	N/A	N/A
2022/0162991	12/2021	Zhang et al.	N/A	N/A
2022/0181859	12/2021	Ji et al.	N/A	N/A
2022/0186724	12/2021	Chang et al.	N/A	N/A
2022/0213777	12/2021	Cui et al.	N/A	N/A
2022/0220836	12/2021	Zhang et al.	N/A	N/A
2022/0224087	12/2021	Ji et al.	N/A	N/A
2022/0228468	12/2021	Cui et al.	N/A	N/A
2022/0228469	12/2021	Zhang et al.	N/A	N/A
2022/0235639	12/2021	Zhang et al.	N/A	N/A
2022/0235640	12/2021	Mao et al.	N/A	N/A
2022/0235641	12/2021	Zhang et al.	N/A	N/A
2022/0235642	12/2021	Zhang et al.	N/A	N/A
2022/0235802	12/2021	Jiang et al.	N/A	N/A
2022/0242297	12/2021	Tian et al.	N/A	N/A
2022/0243613	12/2021	Ji et al.	N/A	N/A
2022/0243724	12/2021	Li et al.	N/A	N/A
2022/0250000	12/2021	Zhang et al.	N/A	N/A
2022/0255319	12/2021	Liu et al.	N/A	N/A
2022/0258659	12/2021	Cui et al.	N/A	N/A
2022/0259947	12/2021	Li et al.	N/A	N/A
2022/0259964	12/2021	Zhang et al.	N/A	N/A
2022/0268201	12/2021	Feng et al.	N/A	N/A
2022/0282606	12/2021	Zhong et al.	N/A	N/A
2022/0282726	12/2021	Zhang et al.	N/A	N/A
2022/0290549	12/2021	Zhang et al.	N/A	N/A
2022/0294194	12/2021	Cao et al.	N/A	N/A
2022/0298906	12/2021	Zhong et al.	N/A	N/A
2022/0307359	12/2021	Liu et al.	N/A	N/A
2022/0307424	12/2021	Wang et al.	N/A	N/A
2022/0314248	12/2021	Ge et al.	N/A	N/A
2022/0315347	12/2021	Liu et al.	N/A	N/A
2022/0316306	12/2021	Liu et al.	N/A	N/A

2022/0316362	12/2021	Zhang et al.	N/A	N/A
2022/0316461	12/2021	Wang et al.	N/A	N/A
2022/0325608	12/2021	Zhang et al.	N/A	N/A
2022/0330411	12/2021	Liu et al.	N/A	N/A
2022/0333471	12/2021	Zhong et al.	N/A	N/A
2022/0339646	12/2021	Yu et al.	N/A	N/A
2022/0341358	12/2021	Ji et al.	N/A	N/A
2022/0341362	12/2021	Feng et al.	N/A	N/A
2022/0341415	12/2021	Deng et al.	N/A	N/A
2022/0345007	12/2021	Liu et al.	N/A	N/A
2022/0349345	12/2021	Zhang et al.	N/A	N/A
2022/0353980	12/2021	Liu et al.	N/A	N/A
2022/0361309	12/2021	Liu et al.	N/A	N/A
2022/0364452	12/2021	Wang et al.	N/A	N/A
2022/0364453	12/2021	Chang et al.	N/A	N/A
2022/0372865	12/2021	Lin et al.	N/A	N/A
2022/0376280	12/2021	Shao et al.	N/A	N/A
2022/0381126	12/2021	Cui et al.	N/A	N/A
2022/0389799	12/2021	Mao	N/A	N/A
2022/0389803	12/2021	Zhang et al.	N/A	N/A
2022/0389804	12/2021	Cui et al.	N/A	N/A
2022/0389865	12/2021	Feng et al.	N/A	N/A
2022/0389867	12/2021	Li et al.	N/A	N/A
2022/0412196	12/2021	Cui et al.	N/A	N/A
2022/0412199	12/2021	Mao et al.	N/A	N/A
2022/0412200	12/2021	Zhang et al.	N/A	N/A
2022/0412258	12/2021	Li et al.	N/A	N/A
2022/0412379	12/2021	Wang et al.	N/A	N/A
2023/0001524	12/2022	Jiang et al.	N/A	N/A
2023/0003238	12/2022	Du et al.	N/A	N/A
2023/0015132	12/2022	Feng et al.	N/A	N/A
2023/0015529	12/2022	Zhang et al.	N/A	N/A
2023/0015581	12/2022	Ji et al.	N/A	N/A
2023/0017968	12/2022	Deng et al.	N/A	N/A
2023/0029574	12/2022	Zhang et al.	N/A	N/A
2023/0029671	12/2022	Han et al.	N/A	N/A
2023/0036118	12/2022	Xing et al.	N/A	N/A
2023/0040970	12/2022	Liu et al.	N/A	N/A
2023/0042379	12/2022	Zhang et al.	N/A	N/A
2023/0047033	12/2022	Fu et al.	N/A	N/A
2023/0048551	12/2022	Feng et al.	N/A	N/A
2023/0049462	12/2022	Zhang et al.	N/A	N/A
2023/0064964	12/2022	Wang et al.	N/A	N/A
2023/0074794	12/2022	Liu et al.	N/A	N/A
2023/0085124	12/2022	Zhong et al.	N/A	N/A
2023/0092506	12/2022	Zhong et al.	N/A	N/A
2023/0092705	12/2022	Liu et al.	N/A	N/A
2023/0106683	12/2022	Zhang et al.	N/A	N/A
2023/0107300	12/2022	Huang et al.	N/A	N/A
2023/0107791	12/2022	Zhang et al.	N/A	N/A
2023/0109018	12/2022	Du et al.	N/A	N/A
2023/0116458	12/2022	Liu et al.	N/A	N/A
2023/0117362	12/2022	Zhang et al.	N/A	N/A
2023/0119725	12/2022	Wang et al.	N/A	N/A
2023/0119876	12/2022	Mao et al.	N/A	N/A
2023/0119896	12/2022	Zhang et al.	N/A	N/A
2023/0120810	12/2022	Fu et al.	N/A	N/A
2023/0121251	12/2022	Cui et al.	N/A	N/A
2023/0124444	12/2022	Chang et al.	N/A	N/A
2023/0138582	12/2022	Li et al.	N/A	N/A
2023/0144116	12/2022	Li et al.	N/A	N/A
2023/0145963	12/2022	Zhang et al.	N/A	N/A
2023/0151722	12/2022	Cui et al.	N/A	N/A
2023/0151723	12/2022	Ji et al.	N/A	N/A
2023/0152793	12/2022	Wang et al.	N/A	N/A
2023/0160289	12/2022	Cui et al.	N/A	N/A
2023/0160510	12/2022	Bao et al.	N/A	N/A
2023/0163580	12/2022	Ji et al.	N/A	N/A
2023/0167776	12/2022	Cui et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
------------	------------------	---------	-----

9609498	12/1998	AU	N/A
737970	12/2000	AU	N/A
2043184	12/1993	CA	N/A
2829762	12/2011	CA	N/A
2737321	12/2012	CA	N/A
2876687	12/2013	CA	N/A
2693567	12/2013	CA	N/A
2964597	12/2016	CA	N/A
2876687	12/2018	CA	N/A
3138533	12/2019	CA	N/A
2919175	12/2020	CA	N/A
2622404	12/2003	CN	N/A
2779054	12/2005	CN	N/A
2890325	12/2006	CN	N/A
200964929	12/2006	CN	N/A
101323151	12/2007	CN	N/A
201190660	12/2008	CN	N/A
201190892	12/2008	CN	N/A
201190893	12/2008	CN	N/A
101414171	12/2008	CN	N/A
201215073	12/2008	CN	N/A
201236650	12/2008	CN	N/A
201275542	12/2008	CN	N/A
201275801	12/2008	CN	N/A
201333385	12/2008	CN	N/A
201443300	12/2009	CN	N/A
201496415	12/2009	CN	N/A
201501365	12/2009	CN	N/A
201507271	12/2009	CN	N/A
101323151	12/2009	CN	N/A
201560210	12/2009	CN	N/A
201581862	12/2009	CN	N/A
201610728	12/2009	CN	N/A
201610751	12/2009	CN	N/A
201618530	12/2009	CN	N/A
201661255	12/2009	CN	N/A
101949382	12/2010	CN	N/A
201756927	12/2010	CN	N/A
101414171	12/2010	CN	N/A
102128011	12/2010	CN	N/A
102140898	12/2010	CN	N/A
102155172	12/2010	CN	N/A
102182904	12/2010	CN	N/A
202000930	12/2010	CN	N/A
202055781	12/2010	CN	N/A
202082265	12/2010	CN	N/A
202100216	12/2011	CN	N/A
202100217	12/2011	CN	N/A
202100815	12/2011	CN	N/A
202124340	12/2011	CN	N/A
202140051	12/2011	CN	N/A
202140080	12/2011	CN	N/A
202144789	12/2011	CN	N/A
202144943	12/2011	CN	N/A
202149354	12/2011	CN	N/A
102383748	12/2011	CN	N/A
202156297	12/2011	CN	N/A
202158355	12/2011	CN	N/A
202163504	12/2011	CN	N/A
202165236	12/2011	CN	N/A
202180866	12/2011	CN	N/A
202181875	12/2011	CN	N/A
202187744	12/2011	CN	N/A
202191854	12/2011	CN	N/A
202250008	12/2011	CN	N/A
101885307	12/2011	CN	N/A
102562020	12/2011	CN	N/A
202326156	12/2011	CN	N/A
202370773	12/2011	CN	N/A
202417397	12/2011	CN	N/A
202417461	12/2011	CN	N/A

102729335	12/2011	CN	N/A
202463955	12/2011	CN	N/A
202463957	12/2011	CN	N/A
202467739	12/2011	CN	N/A
202467801	12/2011	CN	N/A
202531016	12/2011	CN	N/A
202544794	12/2011	CN	N/A
102825039	12/2011	CN	N/A
202578592	12/2011	CN	N/A
202579164	12/2011	CN	N/A
202594808	12/2011	CN	N/A
202594928	12/2011	CN	N/A
202596615	12/2011	CN	N/A
202596616	12/2011	CN	N/A
102849880	12/2012	CN	N/A
102889191	12/2012	CN	N/A
202641535	12/2012	CN	N/A
202645475	12/2012	CN	N/A
202666716	12/2012	CN	N/A
202669645	12/2012	CN	N/A
202669944	12/2012	CN	N/A
202671336	12/2012	CN	N/A
202673269	12/2012	CN	N/A
202751982	12/2012	CN	N/A
102963629	12/2012	CN	N/A
202767964	12/2012	CN	N/A
202789791	12/2012	CN	N/A
202789792	12/2012	CN	N/A
202810717	12/2012	CN	N/A
202827276	12/2012	CN	N/A
202833093	12/2012	CN	N/A
202833370	12/2012	CN	N/A
102140898	12/2012	CN	N/A
202895467	12/2012	CN	N/A
202926404	12/2012	CN	N/A
202935216	12/2012	CN	N/A
202935798	12/2012	CN	N/A
202935816	12/2012	CN	N/A
202970631	12/2012	CN	N/A
103223315	12/2012	CN	N/A
203050598	12/2012	CN	N/A
103233714	12/2012	CN	N/A
103233715	12/2012	CN	N/A
103245523	12/2012	CN	N/A
103247220	12/2012	CN	N/A
103253839	12/2012	CN	N/A
103277290	12/2012	CN	N/A
103321782	12/2012	CN	N/A
203170270	12/2012	CN	N/A
203172509	12/2012	CN	N/A
203175778	12/2012	CN	N/A
203175787	12/2012	CN	N/A
102849880	12/2012	CN	N/A
203241231	12/2012	CN	N/A
203244941	12/2012	CN	N/A
203244942	12/2012	CN	N/A
203303798	12/2012	CN	N/A
102155172	12/2012	CN	N/A
102729335	12/2012	CN	N/A
103420532	12/2012	CN	N/A
203321792	12/2012	CN	N/A
203412658	12/2013	CN	N/A
203420697	12/2013	CN	N/A
203480755	12/2013	CN	N/A
103711437	12/2013	CN	N/A
203531815	12/2013	CN	N/A
203531871	12/2013	CN	N/A
203531883	12/2013	CN	N/A
203556164	12/2013	CN	N/A
203558809	12/2013	CN	N/A
203559861	12/2013	CN	N/A

203559893	12/2013	CN	N/A
203560189	12/2013	CN	N/A
102704870	12/2013	CN	N/A
203611843	12/2013	CN	N/A
203612531	12/2013	CN	N/A
203612843	12/2013	CN	N/A
203614062	12/2013	CN	N/A
203614388	12/2013	CN	N/A
203621045	12/2013	CN	N/A
203621046	12/2013	CN	N/A
203621051	12/2013	CN	N/A
203640993	12/2013	CN	N/A
203655221	12/2013	CN	N/A
103899280	12/2013	CN	N/A
103923670	12/2013	CN	N/A
203685052	12/2013	CN	N/A
203716936	12/2013	CN	N/A
103990410	12/2013	CN	N/A
103993869	12/2013	CN	N/A
203754009	12/2013	CN	N/A
203754025	12/2013	CN	N/A
203754341	12/2013	CN	N/A
203756614	12/2013	CN	N/A
203770264	12/2013	CN	N/A
203784519	12/2013	CN	N/A
203784520	12/2013	CN	N/A
104057864	12/2013	CN	N/A
203819819	12/2013	CN	N/A
203823431	12/2013	CN	N/A
203835337	12/2013	CN	N/A
104074500	12/2013	CN	N/A
203876633	12/2013	CN	N/A
203876636	12/2013	CN	N/A
203877364	12/2013	CN	N/A
203877365	12/2013	CN	N/A
203877375	12/2013	CN	N/A
203877424	12/2013	CN	N/A
203879476	12/2013	CN	N/A
203879479	12/2013	CN	N/A
203890292	12/2013	CN	N/A
203899476	12/2013	CN	N/A
203906206	12/2013	CN	N/A
104150728	12/2013	CN	N/A
104176522	12/2013	CN	N/A
104196464	12/2013	CN	N/A
104234651	12/2013	CN	N/A
203971841	12/2013	CN	N/A
203975450	12/2013	CN	N/A
204020788	12/2013	CN	N/A
204021980	12/2013	CN	N/A
204024625	12/2013	CN	N/A
204051401	12/2013	CN	N/A
204060661	12/2013	CN	N/A
104260672	12/2014	CN	N/A
104314512	12/2014	CN	N/A
204077478	12/2014	CN	N/A
204077526	12/2014	CN	N/A
204078307	12/2014	CN	N/A
204083051	12/2014	CN	N/A
204113168	12/2014	CN	N/A
104340682	12/2014	CN	N/A
104358536	12/2014	CN	N/A
104369687	12/2014	CN	N/A
104402178	12/2014	CN	N/A
104402185	12/2014	CN	N/A
104402186	12/2014	CN	N/A
204209819	12/2014	CN	N/A
204224560	12/2014	CN	N/A
204225813	12/2014	CN	N/A
204225839	12/2014	CN	N/A
104533392	12/2014	CN	N/A

104563938	12/2014	CN	N/A
104563994	12/2014	CN	N/A
104563995	12/2014	CN	N/A
104563998	12/2014	CN	N/A
104564033	12/2014	CN	N/A
204257122	12/2014	CN	N/A
204283610	12/2014	CN	N/A
204283782	12/2014	CN	N/A
204297682	12/2014	CN	N/A
204299810	12/2014	CN	N/A
103223315	12/2014	CN	N/A
104594857	12/2014	CN	N/A
104595493	12/2014	CN	N/A
104612647	12/2014	CN	N/A
104612928	12/2014	CN	N/A
104632126	12/2014	CN	N/A
204325094	12/2014	CN	N/A
204325098	12/2014	CN	N/A
204326983	12/2014	CN	N/A
204326985	12/2014	CN	N/A
204344040	12/2014	CN	N/A
204344095	12/2014	CN	N/A
104727797	12/2014	CN	N/A
204402414	12/2014	CN	N/A
204402423	12/2014	CN	N/A
204402450	12/2014	CN	N/A
103247220	12/2014	CN	N/A
104803568	12/2014	CN	N/A
204436360	12/2014	CN	N/A
204457524	12/2014	CN	N/A
204472485	12/2014	CN	N/A
204473625	12/2014	CN	N/A
204477303	12/2014	CN	N/A
204493095	12/2014	CN	N/A
204493309	12/2014	CN	N/A
103253839	12/2014	CN	N/A
104820372	12/2014	CN	N/A
104832093	12/2014	CN	N/A
104863523	12/2014	CN	N/A
204552723	12/2014	CN	N/A
204553866	12/2014	CN	N/A
204571831	12/2014	CN	N/A
204703814	12/2014	CN	N/A
204703833	12/2014	CN	N/A
204703834	12/2014	CN	N/A
105092401	12/2014	CN	N/A
103233715	12/2014	CN	N/A
103790927	12/2014	CN	N/A
105207097	12/2014	CN	N/A
204831952	12/2014	CN	N/A
204899777	12/2014	CN	N/A
102602323	12/2015	CN	N/A
105240064	12/2015	CN	N/A
204944834	12/2015	CN	N/A
205042127	12/2015	CN	N/A
205172478	12/2015	CN	N/A
103993869	12/2015	CN	N/A
105536299	12/2015	CN	N/A
105545207	12/2015	CN	N/A
205260249	12/2015	CN	N/A
103233714	12/2015	CN	N/A
104340682	12/2015	CN	N/A
205297518	12/2015	CN	N/A
205298447	12/2015	CN	N/A
205391821	12/2015	CN	N/A
205400701	12/2015	CN	N/A
103277290	12/2015	CN	N/A
104260672	12/2015	CN	N/A
205477370	12/2015	CN	N/A
205479153	12/2015	CN	N/A
205503058	12/2015	CN	N/A

205503068	12/2015	CN	N/A
205503089	12/2015	CN	N/A
105958098	12/2015	CN	N/A
205599180	12/2015	CN	N/A
205599180	12/2015	CN	N/A
106121577	12/2015	CN	N/A
205709587	12/2015	CN	N/A
104612928	12/2015	CN	N/A
106246120	12/2015	CN	N/A
205805471	12/2015	CN	N/A
106321045	12/2016	CN	N/A
205858306	12/2016	CN	N/A
106438310	12/2016	CN	N/A
205937833	12/2016	CN	N/A
104563994	12/2016	CN	N/A
206129196	12/2016	CN	N/A
104369687	12/2016	CN	N/A
106715165	12/2016	CN	N/A
106761561	12/2016	CN	N/A
105240064	12/2016	CN	N/A
206237147	12/2016	CN	N/A
206287832	12/2016	CN	N/A
206346711	12/2016	CN	N/A
104563995	12/2016	CN	N/A
107120822	12/2016	CN	N/A
107143298	12/2016	CN	N/A
107159046	12/2016	CN	N/A
107188018	12/2016	CN	N/A
206496016	12/2016	CN	N/A
104564033	12/2016	CN	N/A
107234358	12/2016	CN	N/A
107261975	12/2016	CN	N/A
206581929	12/2016	CN	N/A
104820372	12/2016	CN	N/A
105092401	12/2016	CN	N/A
107476769	12/2016	CN	N/A
107520526	12/2016	CN	N/A
206754664	12/2016	CN	N/A
107605427	12/2017	CN	N/A
106438310	12/2017	CN	N/A
107654196	12/2017	CN	N/A
107656499	12/2017	CN	N/A
107728657	12/2017	CN	N/A
206985503	12/2017	CN	N/A
207017968	12/2017	CN	N/A
107859053	12/2017	CN	N/A
207057867	12/2017	CN	N/A
207085817	12/2017	CN	N/A
105545207	12/2017	CN	N/A
107883091	12/2017	CN	N/A
107902427	12/2017	CN	N/A
107939290	12/2017	CN	N/A
107956708	12/2017	CN	N/A
207169595	12/2017	CN	N/A
207194873	12/2017	CN	N/A
207245674	12/2017	CN	N/A
108034466	12/2017	CN	N/A
108036071	12/2017	CN	N/A
108087050	12/2017	CN	N/A
207380566	12/2017	CN	N/A
108103483	12/2017	CN	N/A
108179046	12/2017	CN	N/A
108254276	12/2017	CN	N/A
108311535	12/2017	CN	N/A
207583576	12/2017	CN	N/A
207634064	12/2017	CN	N/A
207648054	12/2017	CN	N/A
207650621	12/2017	CN	N/A
108371894	12/2017	CN	N/A
207777153	12/2017	CN	N/A
108547601	12/2017	CN	N/A

108547766	12/2017	CN	N/A
108555826	12/2017	CN	N/A
108561098	12/2017	CN	N/A
108561750	12/2017	CN	N/A
108590617	12/2017	CN	N/A
207813495	12/2017	CN	N/A
207814698	12/2017	CN	N/A
207862275	12/2017	CN	N/A
108687954	12/2017	CN	N/A
207935270	12/2017	CN	N/A
207961582	12/2017	CN	N/A
207964530	12/2017	CN	N/A
108789848	12/2017	CN	N/A
108799473	12/2017	CN	N/A
108868675	12/2017	CN	N/A
208086829	12/2017	CN	N/A
208089263	12/2017	CN	N/A
208169068	12/2017	CN	N/A
108979569	12/2017	CN	N/A
109027662	12/2017	CN	N/A
109058092	12/2017	CN	N/A
208179454	12/2017	CN	N/A
208179502	12/2017	CN	N/A
208253147	12/2017	CN	N/A
208260574	12/2017	CN	N/A
109114418	12/2018	CN	N/A
109141990	12/2018	CN	N/A
208313120	12/2018	CN	N/A
208330319	12/2018	CN	N/A
208342730	12/2018	CN	N/A
208430982	12/2018	CN	N/A
208430986	12/2018	CN	N/A
109404274	12/2018	CN	N/A
109429610	12/2018	CN	N/A
109491318	12/2018	CN	N/A
109515177	12/2018	CN	N/A
109526523	12/2018	CN	N/A
109534737	12/2018	CN	N/A
208564504	12/2018	CN	N/A
208564516	12/2018	CN	N/A
208564525	12/2018	CN	N/A
208564918	12/2018	CN	N/A
208576026	12/2018	CN	N/A
208576042	12/2018	CN	N/A
208650818	12/2018	CN	N/A
208669244	12/2018	CN	N/A
109555484	12/2018	CN	N/A
109682881	12/2018	CN	N/A
208730959	12/2018	CN	N/A
208735264	12/2018	CN	N/A
208746733	12/2018	CN	N/A
208749529	12/2018	CN	N/A
208750405	12/2018	CN	N/A
208764658	12/2018	CN	N/A
109736740	12/2018	CN	N/A
109751007	12/2018	CN	N/A
208868428	12/2018	CN	N/A
208870761	12/2018	CN	N/A
109869294	12/2018	CN	N/A
109882144	12/2018	CN	N/A
109882372	12/2018	CN	N/A
209012047	12/2018	CN	N/A
209100025	12/2018	CN	N/A
110080707	12/2018	CN	N/A
110118127	12/2018	CN	N/A
110124574	12/2018	CN	N/A
110145277	12/2018	CN	N/A
110145399	12/2018	CN	N/A
110152552	12/2018	CN	N/A
110155193	12/2018	CN	N/A
110159225	12/2018	CN	N/A

110159432	12/2018	CN	N/A
110159432	12/2018	CN	N/A
110159433	12/2018	CN	N/A
110208100	12/2018	CN	N/A
110252191	12/2018	CN	N/A
110284854	12/2018	CN	N/A
110284972	12/2018	CN	N/A
209387358	12/2018	CN	N/A
110374745	12/2018	CN	N/A
209534736	12/2018	CN	N/A
110425105	12/2018	CN	N/A
110439779	12/2018	CN	N/A
110454285	12/2018	CN	N/A
110454352	12/2018	CN	N/A
110467298	12/2018	CN	N/A
110469312	12/2018	CN	N/A
110469314	12/2018	CN	N/A
110469405	12/2018	CN	N/A
110469654	12/2018	CN	N/A
110485982	12/2018	CN	N/A
110485983	12/2018	CN	N/A
110485984	12/2018	CN	N/A
110486249	12/2018	CN	N/A
110500255	12/2018	CN	N/A
110510771	12/2018	CN	N/A
110513097	12/2018	CN	N/A
209650738	12/2018	CN	N/A
209653968	12/2018	CN	N/A
209654004	12/2018	CN	N/A
209654022	12/2018	CN	N/A
209654128	12/2018	CN	N/A
209656622	12/2018	CN	N/A
107849130	12/2018	CN	N/A
108087050	12/2018	CN	N/A
110566173	12/2018	CN	N/A
110608030	12/2018	CN	N/A
110617187	12/2018	CN	N/A
110617188	12/2018	CN	N/A
110617318	12/2018	CN	N/A
209740823	12/2018	CN	N/A
209780827	12/2018	CN	N/A
209798631	12/2018	CN	N/A
209799942	12/2018	CN	N/A
209800178	12/2018	CN	N/A
209855723	12/2018	CN	N/A
209855742	12/2018	CN	N/A
209875063	12/2018	CN	N/A
110656919	12/2019	CN	N/A
107520526	12/2019	CN	N/A
110787667	12/2019	CN	N/A
110821464	12/2019	CN	N/A
110833665	12/2019	CN	N/A
110848028	12/2019	CN	N/A
210049880	12/2019	CN	N/A
210049882	12/2019	CN	N/A
210097596	12/2019	CN	N/A
210105817	12/2019	CN	N/A
210105818	12/2019	CN	N/A
210105993	12/2019	CN	N/A
110873093	12/2019	CN	N/A
210139911	12/2019	CN	N/A
110947681	12/2019	CN	N/A
111058810	12/2019	CN	N/A
111075391	12/2019	CN	N/A
210289931	12/2019	CN	N/A
210289932	12/2019	CN	N/A
210289933	12/2019	CN	N/A
210303516	12/2019	CN	N/A
211412945	12/2019	CN	N/A
111089003	12/2019	CN	N/A
111151186	12/2019	CN	N/A

111167769	12/2019	CN	N/A
111169833	12/2019	CN	N/A
111173476	12/2019	CN	N/A
111185460	12/2019	CN	N/A
111185461	12/2019	CN	N/A
111188763	12/2019	CN	N/A
111206901	12/2019	CN	N/A
111206992	12/2019	CN	N/A
111206994	12/2019	CN	N/A
210449044	12/2019	CN	N/A
210460875	12/2019	CN	N/A
210522432	12/2019	CN	N/A
210598943	12/2019	CN	N/A
210598945	12/2019	CN	N/A
210598946	12/2019	CN	N/A
210599194	12/2019	CN	N/A
210599303	12/2019	CN	N/A
210600110	12/2019	CN	N/A
111219326	12/2019	CN	N/A
111350595	12/2019	CN	N/A
210660319	12/2019	CN	N/A
210714569	12/2019	CN	N/A
210769168	12/2019	CN	N/A
210769169	12/2019	CN	N/A
210769170	12/2019	CN	N/A
210770133	12/2019	CN	N/A
210825844	12/2019	CN	N/A
210888904	12/2019	CN	N/A
210888905	12/2019	CN	N/A
210889242	12/2019	CN	N/A
111397474	12/2019	CN	N/A
111412064	12/2019	CN	N/A
111441923	12/2019	CN	N/A
111441925	12/2019	CN	N/A
111503517	12/2019	CN	N/A
111515898	12/2019	CN	N/A
111594059	12/2019	CN	N/A
111594062	12/2019	CN	N/A
111594144	12/2019	CN	N/A
211201919	12/2019	CN	N/A
211201920	12/2019	CN	N/A
211202218	12/2019	CN	N/A
111608965	12/2019	CN	N/A
111664087	12/2019	CN	N/A
111677476	12/2019	CN	N/A
111677647	12/2019	CN	N/A
111692064	12/2019	CN	N/A
111692065	12/2019	CN	N/A
211384571	12/2019	CN	N/A
211397553	12/2019	CN	N/A
211397677	12/2019	CN	N/A
211500955	12/2019	CN	N/A
211524765	12/2019	CN	N/A
4004854	12/1990	DE	N/A
4241614	12/1993	DE	N/A
102009022859	12/2009	DE	N/A
102012018825	12/2013	DE	N/A
102013111655	12/2013	DE	N/A
102015103872	12/2014	DE	N/A
102013114335	12/2019	DE	N/A
0835983	12/1997	EP	N/A
1378683	12/2003	EP	N/A
2143916	12/2009	EP	N/A
2613023	12/2012	EP	N/A
3095989	12/2015	EP	N/A
3211766	12/2016	EP	N/A
3049642	12/2017	EP	N/A
3354866	12/2017	EP	N/A
3075946	12/2018	EP	N/A
2795774	12/1998	FR	N/A
474072	12/1936	GB	N/A

1438172	12/1975	GB	N/A
857135212	12/1983	JP	N/A
20020026398	12/2001	KR	N/A
13562	12/1999	RU	N/A
1993020328	12/1992	WO	N/A
2006025886	12/2005	WO	N/A
2009023042	12/2008	WO	N/A
20110133821	12/2010	WO	N/A
2012139380	12/2011	WO	N/A
2013158822	12/2012	WO	N/A
PCT/CN2012/074945	12/2012	WO	N/A
2013185399	12/2012	WO	N/A
2015158020	12/2014	WO	N/A
2016014476	12/2015	WO	N/A
2016033983	12/2015	WO	N/A
2016078181	12/2015	WO	N/A
2016101374	12/2015	WO	N/A
2016112590	12/2015	WO	N/A
2016/186790	12/2015	WO	N/A
2017123656	12/2016	WO	N/A
2017146279	12/2016	WO	N/A
2017213848	12/2016	WO	N/A
2018031029	12/2017	WO	N/A
2018038710	12/2017	WO	N/A
2018044293	12/2017	WO	N/A
2018044307	12/2017	WO	N/A
2018071738	12/2017	WO	N/A
2018101909	12/2017	WO	N/A
2018101912	12/2017	WO	N/A
2018106210	12/2017	WO	N/A
2018106225	12/2017	WO	N/A
2018106252	12/2017	WO	N/A
2018/132106	12/2017	WO	N/A
2018156131	12/2017	WO	N/A
2018075034	12/2017	WO	N/A
2018187346	12/2017	WO	N/A
2018031031	12/2018	WO	N/A
2019045691	12/2018	WO	N/A
2019046680	12/2018	WO	N/A
2019060922	12/2018	WO	N/A
2019117862	12/2018	WO	N/A
2019126742	12/2018	WO	N/A
2019147601	12/2018	WO	N/A
2019169366	12/2018	WO	N/A
2019195651	12/2018	WO	N/A
2019200510	12/2018	WO	N/A
2019210417	12/2018	WO	N/A
2020018068	12/2019	WO	N/A
2020046866	12/2019	WO	N/A
2020072076	12/2019	WO	N/A
2020076569	12/2019	WO	N/A
2020097060	12/2019	WO	N/A
2020104088	12/2019	WO	N/A
2020131085	12/2019	WO	N/A
2020211083	12/2019	WO	N/A
2020211086	12/2019	WO	N/A
2021038604	12/2020	WO	N/A
2021041783	12/2020	WO	N/A

OTHER PUBLICATIONS

Rigmaster Machinery Ltd., Model: 2000 RMP-6-PLEX, brochure, downloaded at https://www.rigmastermachinery.com/_files/ugd/431e62_eaecd77c9fe54af8b13d08396072da67.pdf. cited by applicant

Final written decision of PGR2021-00102 dated Feb. 6, 2023. cited by applicant

Final written decision of PGR2021-00103 dated Feb. 6, 2023. cited by applicant

AFGlobal Corporation, Durastim Hydraulic Fracturing Pump, A Revolutionary Design for Continuous Duty Hydraulic Fracturing, 2018. cited by applicant

SPM® QEM 5000 E-Frac Pump Specification Sheet, Weir Group (2019) (“Weir 5000”). cited by applicant

Green Field Energy Services Natural Gas Driven Turbine Frac Pumps HHP Summit Presentation, Yumpu (Sep. 2012), <https://www.yumpu.com/en/document/read/49685291/turbine-frac-pump-assembly-hhp> (“Green Field”). cited by applicant

Dowell B908 “Turbo-Jet” Operator’s Manual. cited by applicant

Jereh Debut’s Super-power Turbine Fracturing Pump, Leading the Industrial Revolution, Jereh Oilfield Services Group (Mar. 19, 2014),

<https://www.prnewswire.com/news-releases/jereh-debuts-super-power-turbine-fracturing-pump-leading-the-industrial-revolution-250992111.html>. cited by applicant

Jereh Apollo 4500 Turbine Frac Pumper Finishes Successful Field Operation in China, Jereh Group (Feb. 13, 2015), as available on Apr. 20, 2015, <https://web.archive.org/web/20150420220625/https://www.prnewswire.com/news-releases/jereh-apollo-4500-turbine-frac-pumper-finishes-successful-field-operation-in-china-300035829.html>. cited by applicant

35% Economy Increase, Dual-fuel System Highlighting Jereh Apollo Frac Pumper, Jereh Group (Apr. 13, 2015), <https://www.jereh.com/en/news/press-release/news-detail-7345.htm>. cited by applicant

Hydraulic Fracturing: Gas turbine proves successful in shale gas field operations, Vericor (2017), <https://www.vericor.com/wp-content/uploads/2020/02/7.-Fracing-4500hp-Pump-China-En.pdf> ("Vericor Case Study"). cited by applicant

Jereh Apollo Turbine Fracturing Pumper Featured on China Central Television, Jereh Group (Mar. 9, 2018), <https://www.jereh.com/en/news/press-release/news-detail-7267.htm>. cited by applicant

Jereh Unveiled New Electric Fracturing Solution at OTC 2019, Jereh Group (May 7, 2019), as available on May 28, 2019, <https://web.archive.org/web/20190528183906/https://www.prnewswire.com/news-releases/jereh-unveiled-new-electric-fracturing-solution-at-otc-2019-300845028.html>. cited by applicant

Jereh Group, Jereh Fracturing Unit, Fracturing Spread, YouTube (Mar. 30, 2015), <https://www.youtube.com/watch?v=PIkDbU5dEOo>. cited by applicant

Transcript of Jereh Group, Jereh Fracturing Unit, Fracturing Spread, YouTube (Mar. 30, 2015). cited by applicant

Jereh Group, Jereh Fracturing Equipment. YouTube (Jun. 8, 2015), <https://www.youtube.com/watch?v=m0vMiq84P4Q>. cited by applicant

Transcript of Jereh Group, Jereh Fracturing Equipment, YouTube (Jun. 8, 2015), <https://www.youtube.com/watch?v=m0vMiq84P4Q>. cited by applicant

Ferdinand P. Beer et al., *Mechanics of Materials* (6th ed. 2012). cited by applicant

Weir Oil & Gas Introduces Industry's First Continuous Duty 5000-Horsepower Pump, Weir Group (Jul. 25, 2019), <https://www.global.weir/newsroom/news-articles/weir-oil-and-gas-introduces-industrys-first-continuous-duty-5000-horsepower-pump/>. cited by applicant

2012 High Horsepower Summit Agenda, Natural Gas for High Horsepower Applications (Sep. 5, 2012). cited by applicant

Review of HHP Summit 2012, Gladstein, Neandross & Associates <https://www.gladstein.org/gna-conferences/high-horsepower-summit-2012/>. cited by applicant

Green Field Energy Services Deploys Third New Hydraulic Fracturing System, Green Field Energy Services, Inc. (Jul. 11, 2012), <https://www.prnewswire.com/news-releases/green-field-energy-services-deploys-third-new-hydraulic-fracturing-spread-162113425>. cited by applicant

Karen Boman, Turbine Technology Powers Green Field Multi-Fuel Frack Pump, Rigzone (Mar. 7, 2015), as available on Mar. 14, 2015, https://web.archive.org/web/20150314203227/https://www.rigzone.com/news/oil-gas/a/124883/Turbine_Technology_Powers_Green_Field_MultiFuel_Frack_Pump. cited by applicant

"Turbine Frac Units," WMD Squared (2012), <https://wmdsquared.com/work/gfes-turbine-frac-units/>. cited by applicant

Leslie Turj, Green Field asset sale called 'largest disposition industry has seen,' The INDsider Media (Mar. 19, 2014), <http://theind.com/article-16497-green-field-asset-sale-called-%E2%80%98largest-disposition-industry-has-seen%60.html>. cited by applicant

"Honghua developing new-generation shale-drilling rig, plans testing of frac pump"; Katherine Scott; Drilling Contractor May 23, 2013; accessed at <https://www.drillingcontractor.org/honghua-developing-new-generation-shale-drilling-rig-plans-testing-of-frac-pump-23278>. cited by applicant

Europump and Hydrualic Institute, *Variable Speed Pumping: A Guide to Successful Applications*, Elsevier Ltd, 2004. cited by applicant

Capstone Turbine Corporation, Capstone Receives Three Megawatt Order from Large Independent Oil & Gas Company in Eagle Ford Shale Play, Dec. 7, 2010. cited by applicant

Wikipedia, Westinghouse Combustion Turbine Systems Division, https://en.wikipedia.org/wiki/Westinghouse_Combustion_Turbine_Systems_Division, circa 1960. cited by applicant

Wikipedia, Union Pacific GTELs, https://en.wikipedia.org/wiki/Union_Pacific_GTEs, circa 1950. cited by applicant

HCI JET Frac, Screenshots from YouTube, Dec. 11, 2010. <https://www.youtube.com/watch?v=6HjXkdbFaFQ>. cited by applicant

Afd Petroleum Ltd., Automated Hot Zone, Frac Refueling System, Dec. 2018. cited by applicant

Eygun, Christiane, et al., URTEc: 2687987, Mitigating Shale Gas Developments Carbon Footprint: Evaluating and Implementing Solutions in Argentina, Copyright 2017, Unconventional Resources Technology Conference. cited by applicant

Walzel, Brian, Hart Energy, Oil, Gas Industry Discovers Innovative Solutions to Environmental Concerns, Dec. 10, 2018. cited by applicant

Frac Shack, Bi-Fuel FracFueller brochure, 2011. cited by applicant

Pettigrew, Dana, et al., High Pressure Multi-Stage Centrifugal Pump for 10,000 psi Frac Pump—HPHPS FRAC Pump, Copyright 2013, Society of Petroleum Engineers, SPE 166191. cited by applicant

Elle Seybold, et al., Evolution of Dual Fuel Pressure Pumping for Fracturing: Methods, Economics, Field Trial Results and Improvements in Availability of Fuel, Copyright 2013, Society of Petroleum Engineers, SPE 166443. cited by applicant

Wallace, E.M., Associated Shale Gas: From Flares to Rig Power, Copyright 2015, Society of Petroleum Engineers, SPE-173491-MS. cited by applicant

Williams, C.W. (Gulf Oil Corp. Odessa Texas), The Use of Gas-turbine Engines in an Automated High-Pressure Water-injection Stations; American Petroleum Institute; API-63-144 (Jan. 1, 1963). cited by applicant

Neal, J.C. (Gulf Oil Corp. Odessa Texas), Gas Turbine Driven Centrifugal Pumps for High Pressure Water Injection; American Institute of Mining, Metallurgical and Petroleum Engineers, Inc.; SPE-1888 (1967). cited by applicant

Porter, John A. (SOLAR Division International Harvester Co.), Modern Industrial Gas Turbines for the Oil Field; American Petroleum Institute; Drilling and Production Practice; API-67-243 (Jan. 1, 1967). cited by applicant

Cooper et al., Jet Frac Porta-Skid—A New Concept in Oil Field Service Pump Equipments[sic]; Halliburton Services; SPE-2706 (1969). cited by applicant

Ibragimov, É.S., Use of gas-turbine engines in oil field pumping units; Chem Petrol Eng; (1994) 30: 530. <https://doi.org/10.1007/BF01154919>. (Translated from Khimicheskaya i Neftyanoe Mashinostroenie, No. 11, pp. 24-26, Nov. 1994.). cited by applicant

Kas'yanov et al., Application of gas-turbine engines in pumping units complexes of hydraulic fracturing of oil and gas reservoirs; Exposition Oil & Gas (Oct. 2012) (published in Russian). cited by applicant

American Petroleum Institute. API 674: Positive Displacement Pumps—Reciprocating. 3rd ed. Washington, DC: API Publishing Services, 2010. cited by applicant

American Petroleum Institute. API 616: Gas Turbines for the Petroleum, Chemical, and Gas Industry Services. 5th ed. Washington, DC: API Publishing Services, 2011. cited by applicant

Karassik, Igor, Joseph Messina, Paul Cooper, and Charles Heald. Pump Handbook. 4th ed. New York: McGraw-Hill Education, 2008. cited by applicant

Weir SPM. Weir SPM General Catalog: Well Service Pumps, Flow Control Products, Manifold Trailers, Safety Products, Post Sale Services. Ft. Worth TX: Weir Oil & Gas. May 28, 2016. <https://www.pumpfundamentals.com/pumpdatabase2/weir-spm-general.pdf>. cited by applicant

The Group, Inc. Weir SPM Pump Product Catalog. Ft. Worth, TX: S.P.M. Flow Control, Inc. Oct. 30, 2017. https://manage.global.weir/assets/files/product%20brochures/SPM_2P140706_Pump_Product_Catalogue_View.pdf. cited by applicant

Shandong Saigao Group Corporation. Q4 (5W115) Quintuplex Plunger Pump. Jinan City, Shandong Province, China: Saigao. Oct. 20, 2014. <https://www.saigaogroup.com/product/q400-5w115-quintuplex-plunger-pump.html>. cited by applicant

Marine Turbine. Turbine Powered Frac Units. Franklin, Louisiana: Marine Turbine Technologies, 2020. cited by applicant

Rotating Right. Quintuplex Power Pump Model Q700. Edmonton, Alberta, Canada: Weatherford International Ltd. <https://www.rotatingright.com/pdf/weatherford/RR%2026-Weatherford%20Model%20Q700.pdf>, 2021. cited by applicant

CanDyne Pump Services, Inc. Weatherford Q700 Pump. Calgary, Alberta, Canada: CanDyne Pump Services. Aug. 15, 2015. <http://candyne.com/wp-content/uploads/2014/10/181905-94921.q700-quintuplex-pump.pdf>. cited by applicant

Arop, Julius Bankong. Geomechanical review of hydraulic fracturing technology. Thesis (M. Eng.). Cambridge, MA: Massachusetts Institute of Technology, Dept. of Civil and Environmental Engineering. Oct. 29, 2013. <https://dspace.mit.edu/handle/1721.1/82176>. cited by applicant

ResearchGate, Answer by Byron Woolridge, found at https://www.researchgate.net/post/How_can_we_improve_the_efficiency_of_the_gas_turbine_cycles, Jan. 1, 2013. cited by applicant

Filipović, Ivan, Preliminary Selection of Basic Parameters of Different Torsional Vibration Dampers Intended for use in Medium-Speed Diesel Engines. Transactions of Fama XXXVI-3 (2012). cited by applicant

Marine Turbine Technologies, 1 MW Power Generation Package, <http://marineturbine.com/power-generation>, 2017. cited by applicant

Business Week: Fiber-optic cables help fracking, cablinginstall.com. Jul. 12, 2013. <https://www.cablinginstall.com/cable/article/16474208/businessweek-fiberoptic-cables-help-fracking>. cited by applicant

Fracking companies switch to electric motors to power pumps, iadd-intl.org. Jun. 27, 2019. <https://www.iadd-intl.org/articles/fracking-companies-switch-to-electric-motors-to-power-pumps/>. cited by applicant

The Leader in Frac Fueling, suncoastresources.com. Jun. 29, 2015. <https://web.archive.org/web/20150629220609/https://www.suncoastresources.com/oilfield/fueling-services/>. cited by applicant

Mobile Fuel Delivery, atlasoil.com. Mar. 6, 2019. <https://www.atlasoil.com/nationwide-fueling/onsite-and-mobile-fueling>. cited by applicant

Frac Tank Hose (FRAC), 4starhose.com. Accessed: Nov. 10, 2019. http://www.4starhose.com/product/frac_tank_hose_frac.aspx. cited by applicant

Plos One, Dynamic Behavior of Reciprocating Plunger Pump Discharge Valve Based on Fluid Structure Interaction and Experimental Analysis. Oct. 2015. cited by applicant

FMC Technologies, Operation and Maintenance Manual, L06 Through L16 Triplex Pumps Doc No. OMM50000903 Rev: E p. 1 of 66. Aug. 27, 2009. cited by applicant

Gardner Denver Hydraulic Fracturing Pumps GD 3000 <https://www.gardnerdenver.com/en-us/pumps/triplex-fracking-pump-gd-3000>. cited by applicant

Lekontsev, Yu M., et al. "Two-side sealer operation." Journal of Mining Science 49.5 (2013): 757-762. cited by applicant

Tom Hausfeld, GE Power & Water, and Eldon Schelske, Evolution Well Services, TM2500+ Power for Hydraulic Fracturing. cited by applicant

FTS International's Dual Fuel Hydraulic Fracturing Equipment Increases Operational Efficiencies, Provides Cost Benefits, Jan. 3, 2018. cited by applicant

CNG Delivery, Fracturing with natural gas, dual-fuel drilling with CNG, Aug. 22, 2019. cited by applicant

PbNG, Natural Gas Fuel for Drilling and Hydraulic Fracturing, Diesel Displacement / Dual Fuel & Bi-Fuel, May 2014. cited by applicant

Integrated Flow, Skid-mounted Modular Process Systems, Jul. 15, 2017, <https://ifsolutions.com/why-modular/>. cited by applicant

Cameron, A Schlumberger Company, Frac Manifold Systems, 2016. cited by applicant

ZSI-Foster, Energy | Solar | Fracking | Oil and Gas, Aug. 2020, <https://www.zsi-foster.com/energy-solar-fracking-oil-and-gas.html>. cited by applicant

JBG Enterprises, Inc., WS-Series Blowout Prevention Safety Coupling—Quick Release Couplings, Sep. 11, 2015, <http://www.jgbhose.com/products/WS-Series-Blowout-Prevention-Safety-Coupling.asp>. cited by applicant

Halliburton, Vessel-based Modular Solution (VMS), 2015. cited by applicant

Chun, M. K., H. K. Song, and R. Lallemand. "Heavy duty gas turbines in petrochemical plants: Samsung's Daesan plant (Korea) beats fuel flexibility records with over 95% hydrogen in process gas." Proceedings of PowerGen Asia Conference, Singapore. 1999. cited by applicant

Wolf, Jürgen J., and Marko A. Perkavec. "Safety Aspects and Environmental Considerations for a 10 MW Cogeneration Heavy Duty Gas Turbine Burning Coke Oven Gas with 60% Hydrogen Content." ASME 1992 International Gas Turbine and Aeroengine Congress and Exposition. American Society of Mechanical Engineers Digital Collection, 1992. cited by applicant

Ginter, Timothy, and Thomas Bouvay. "Uprate options for the MS7001 heavy duty gas turbine." GE paper GER-3808C, GE Energy 12 (2006). cited by applicant

Chaichan, Miqdam Tariq. "The impact of equivalence ratio on performance and emissions of a hydrogen-diesel dual fuel engine with cooled exhaust gas recirculation." International Journal of Scientific & Engineering Research 6.6 (2015): 938-941. cited by applicant

Ecob, David J., et al. "Design and Development of a Landfill Gas Combustion System for the Typhoon Gas Turbine." ASME 1996 International Gas Turbine and Aeroengine Congress and Exhibition. American Society of Mechanical Engineers Digital Collection, 1996. cited by applicant

II-VI Marlow Industries, Thermoelectric Technologies in Oil, Gas, and Mining Industries, blog.marlow.com (Jul. 24, 2019). cited by applicant

B.M. Mahlalela, et al., "Electric Power Generation Potential Based on Waste Heat and Geothermal Resources in South Africa," pangea.stanford.edu (Feb. 11, 2019). cited by applicant

Department of Energy, United States of America, The Water-Energy Nexus: Challenges and Opportunities ourenergypolicy.org (Jun. 2014). cited by applicant

Ankit Tiwari, Design of a Cooling System for a Hydraulic Fracturing Equipment, The Pennsylvania State University, The Graduate School, College of Engineering, 2015. cited by applicant

Jp Yadav et al., Power Enhancement of Gas Turbine Plant by Intake Air Fog Cooling, Jun. 2015. cited by applicant

Mee Industries: Inlet Air Fogging Systems for Oil, Gas and Petrochemical Processing, Verdict Media Limited Copyright 2020. cited by applicant

M. Ahmadzadehtalatapeh et al. Performance enhancement of gas turbine units by retrofitting with inlet air cooling technologies (IACs): an hour-by-hour simulation study, Journal of the Brazilian Society of Mechanical Sciences and Engineering, Mar. 2020. cited by applicant

Advances in Popular Torque-Link Solution Offer OEMs Greater Benefit, Jun. 21, 2018. cited by applicant

Emmanuel Akita et al., Mewbourne College of Earth & Energy, Society of Petroleum Engineers; Drilling Systems Automation Technical Section (DSATS); 2019. cited by applicant

PowerShelter Kit II, nooutage.com, Sep. 6, 2019. cited by applicant

EMPengineering.com, HEMP Resistant Electrical Generators / Hardened Structures HEMP/GMD Shielded Generators, Virginia, Nov. 3, 2012. cited by applicant

Blago Minovski, Coupled Simulations of Cooling and Engine Systems for Unsteady Analysis of the Benefits of Thermal Engine Encapsulation, Department of Applied Mechanics, Chalmers University of Technology G-öteborg, Sweden 2015. cited by applicant

J. Porteiro et al., Feasibility of a new domestic CHP trigeneration with heat pump: II. Availability analysis. Design and development, Applied Thermal Engineering 24 (2004) 1421-1429. cited by applicant

ISM, What is Cracking Pressure, 2019. cited by applicant

Swagelok, The right valve for controlling flow direction? Check, 2016. cited by applicant

Technology.org, Check valves how do they work and what are the main type, 2018. cited by applicant

De Gevigney et al., "Analysis of no-load dependent power losses in a planetary gear train by using thermal network method", International Gear Conference 2014: Aug. 26-28, 2014, Lyon, pp. 615-624. cited by applicant

Special-Purpose Couplings for Petroleum, Chemical, and Gas Industry Services, API Standard 671 (4th Edition) (2010). cited by applicant

The Application of Flexible Couplings for Turbomachinery, Jon R. Mancuso et al., Proceedings of the Eighteenth turbomachinery Symposium (1989). cited by applicant

Pump Control With Variable Frequency Drives, Kevin Tory, Pumps & Systems: Advances in Motors and Drives, Reprint from Jun. 2008. cited by applicant

Fracture Design and Stimulation, Mike Eberhard, P.E., Wellconstruction & Operations Technical Workshop Insupport of the Epa Hydraulic Fracturing Study, Mar. 10-11, 2011. cited by applicant

General Purpose vs. Special Purpose Couplings, Jon Mancuso, Proceedings of the Twenty-Third Turbomachinerysymposium (1994). cited by applicant

Overview of Industry Guidance/Best Practices on Hydraulic Fracturing (HF), American Petroleum Institute, © 2012. cited by applicant

API Member Companies, American Petroleum Institute, WaybackMachine Capture, <https://web.archive.org/web/20130424080625/http://api.org/globalitems/globalheaderpages/membership/api-member-companies>, accessed Jan. 4, 2021. cited by applicant

API's Global Industry Services, American Petroleum Institute, © Aug. 2020. cited by applicant

About API, American Petroleum Institute, <https://www.api.org/about>, accessed Dec. 30, 2021. cited by applicant

About API, American Petroleum Institute, WaybackMachine Capture, <https://web.archive.org/web/20110422104346/http://api.org/aboutapi/>, captured Apr. 22, 2011. cited by applicant

Publications, American Petroleum Institute, WaybackMachine Capture, <https://web.archive.org/web/20110427043936/http://www.api.org:80/Publications/>, captured Apr. 27, 2011. cited by applicant

Procedures for Standards Development, American Petroleum Institute, Third Edition (2006). cited by applicant

WorldCat Library Collections Database Records for API Standard 671 and API Standard 674, https://www.worldcat.org/title/positive-displacement-pumps-reciprocating/oclc/858692269&referer=brief_results, accessed Dec. 30, 2021; and https://www.worldcat.org/title/special-purpose-couplings-for-petroleum-chemical-and-gas-industry-services/oclc/871254217&referer=brief_results, accessed Dec. 22, 2021. cited by applicant

2011 Publications and Services, American Petroleum Institute (2011). cited by applicant

Standards, American Petroleum Institute, WaybackMachine Capture, <https://web.archive.org/web/20110207195046/http://www.api.org/Standards/>, captured Feb. 7, 2011; and <https://web.archive.org/web/20110204112554/http://global.ihs.com/?RID=API1>, captured Feb. 4, 2011. cited by applicant

IHS Markit Standards Store, https://global.ihs.com/doc_detail.cfm?document_name=API%20STD%20674&item_s_key=00010672#doc-detail-history, accessed Dec. 30, 2021; and https://global.ihs.com/doc_detail.cfm?document_name=API%20STD%20671&item_s_key=00010669&item_key_date=890331&origin=DSS&input_doc_number=671&input_doc_title=&document_name=API%20STD%20671&item_s_key=00010669&item_key_date=890331&origin=DSS, accessed Dec. 30, 2021. cited by applicant

Dziubak, Tadeusz, "Experimental Studies of Dust Suction Irregularity from Multi-Cyclone Dust Collector of Two-Stage Air Filter", Energies 2021, 14, 3577, 28 pages. cited by applicant

Primary Examiner: Buttar; Mandeep S

Attorney, Agent or Firm: Norton Rose Fulbright US LLP

Background/Summary

PRIORITY CLAIM (1) This is a continuation of U.S. Non-Provisional application Ser. No. 18/096,927, filed Jan. 13, 2023, titled "ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT AND RELATED METHODS," which is a continuation of U.S. Non-Provisional application Ser. No. 17/444,485, filed Aug. 5, 2021, titled "ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT AND RELATED METHODS," now U.S. Pat. No. 11,627,683, issued Apr. 11, 2023, which is a continuation of U.S. Non-Provisional application Ser. No. 17/356,063, filed Jun. 23, 2021, titled "ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT AND RELATED METHODS," now U.S. Pat. No. 11,129,295, issued Sep. 21, 2021, which is a divisional of U.S. Non-Provisional application Ser. No. 17/302,039, filed Apr. 22, 2021, titled "ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT AND RELATED METHODS," now U.S. Pat. No. 11,109,508, issued Aug. 31, 2021, which claims priority to and the benefit of, under 35 U.S.C. § 119(e), U.S. Provisional Application No. 62/705,042, filed Jun. 9, 2020, titled "ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT AND RELATED METHODS," and U.S. Provisional Application No. 62/704,981, filed Jun. 5, 2020, titled "ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT (DDU) AND RELATED METHODS," the disclosures of which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

(1) This disclosure relates to enclosure assemblies and related systems and methods for providing enhanced cooling of a direct drive unit (DDU), such as a direct drive turbine (DDT) connected to a gearbox for driving a driveshaft connected to a pump for use in a hydraulic fracturing systems and methods.

BACKGROUND

(2) During fracturing operations, the equipment onboard fracturing trailers utilizes extensive cooling to facilitate operation throughout the pumping stage. The fracturing pump may have, for example, up to 5% energy loss of energy through heat rejection during operation. Such heat rejection may enter bearings, connecting rods, the casing, clamps and other highly temperature sensitive components in the pumps power end. These components are typically kept lubricated and cooled using lube oil that is pumped continuously through circuits into the pump ensuring that the lube oil is cascaded around the crank case of the fluid pump.

(3) Heat rejection from the pump is still absorbed into the oil, however, and this oil is cooled through a lubrication circuit to ensure that the oil remains at a manageable temperature set out by regulation and/or pump original equipment manufacturers (OEMs). The cooling of oil may be

achieved by diverting the oil to a heat exchanger (for example, a fan driven heat exchanger, tube and shell heat exchanger, or other heat exchanger as will be understood by those skilled in the art.) that is sized and configured to be able to remove enough heat from the fluid that will allow the oil to enter the crank case again and absorb more heat rejection.

(4) This cooling cycle may occur constantly onboard fracturing trailers with the operations of the heat exchangers at times being hydraulically or electrically driven. The need for higher power rated fracturing pumps, for example, 5000 HP or 7000 HP rated fracturing pumps, may require larger cooling packages to be able to manage the heat rejection. Accordingly, more heat rejection may directly correlate to the physical footprint of the cooling systems.

SUMMARY OF THE DISCLOSURE

(5) In view of the foregoing, there is an ongoing need for an enclosure assembly and related systems and methods that are more suitable for cooling the DDU of a pumping system, as well as for high-pressure and high-power operations.

(6) Accordingly, it may be seen that a need exists for managing the location of cooling systems to minimize physical footprint, for managing associated power resources efficiently, and for providing effective cooling for fracturing pumps and DDUs. The present disclosure addresses these and other related and unrelated problems in the art.

(7) One exemplary embodiment of the disclosure includes an enclosure assembly to enhance cooling of a hydraulic fracturing direct drive unit (DDU) during operation. An enclosure body may be provided extending at least partially around an enclosure space to house the DDU, which may include a turbine engine that is mechanically connected to a gearbox for driving a driveshaft connected to the gearbox in order to drive a fluid pump. The enclosure assembly may include one or more heat exchanger assemblies connected to the enclosure body for cooling a process fluid associated with one or more of the DDU and the fluid pump, for example, a lubrication or other lubrication medium, and/or a hydraulic/working fluid that is heated during operation. The one or more heat exchanger assemblies may include one or more intake fan assemblies positioned in fluid communication with an external environment surrounding the enclosure body, and one or more intake fan motors may be operatively connected to the one or more intake fan assemblies. Thus, when the one or more intake fan motors is activated, the one or more intake fan assemblies may draw air into the enclosure space from the external environment at the one or more intake fan assemblies and along an airflow path through the enclosure space. One or more radiator assemblies may further be included in the one or more heat exchanger assemblies for receiving the process fluid, and positioned adjacent the one or more intake fan assemblies in the airflow path through the enclosure space to cool the process fluid with air from the external environment as it flows toward the radiator assembly.

(8) In addition, the enclosure assembly may include one or more outlet fan assemblies positioned in fluid communication with the external environment. Accordingly, to maintain a desired temperature of the enclosure space, the one or more outlet fan assemblies may be operatively connected to one or more outlet fan motors to discharge air from the enclosure space to the external environment when the one or more outlet fan motors is activated such that airflow heated by the cooling of the process fluid may be ventilated from the enclosure assembly. The enclosure assembly may also include one or more temperature sensors to detect a temperature of the enclosure space and, further, one or more controllers in electrical communication with the one or more temperature sensors. The one or more controllers may be operatively connected to one or more of the one or more intake fan motors and the one or more outlet fan motors. In this regard, the one or more controllers may activate the respective one or more intake fan motors and the one or more outlet fan motors to rotate the respective one or more intake fan assemblies and the one or more outlet fan assemblies responsive to a predetermined temperature signal from the one or more temperature sensors to discharge heated air from and maintain a desired temperature of the enclosure space.

(9) Another exemplary embodiment of the disclosure includes a fluid pumping system for high-pressure, high-power hydraulic fracturing operations. The system may include a direct drive unit (DDU) having a turbine engine mechanically connected to a gearbox for driving a driveshaft, and a fluid pump operatively connected to the DDU by the driveshaft for driving the fluid pump. Accordingly, one or more of the DDU and the fluid pump may generate and heat process fluid during operation, which may include lubrication oil or another lubrication medium, and/or a hydraulic or other working fluid. The system may include an enclosure assembly having an enclosure body extending around an enclosure space to house the DDU, and one or more heat exchanger assemblies connected to the enclosure body for cooling process fluid associated with one or more of the DDU and the fluid pump. The one or more heat exchanger assemblies of the system may include one or more intake fan assemblies positioned in fluid communication with an external environment surrounding the enclosure body, and one or more intake fan motors may be operatively connected to the one or more intake fan assemblies. When the one or more intake fan motors is activated, the one or more intake fan assemblies may draw air into the enclosure space from the external environment at the one or more intake fan assemblies and along an airflow path through the enclosure space. One or more radiator assemblies may be included in the one or more heat exchanger assemblies for receiving the process fluid, and may be positioned adjacent the one or more intake fan assemblies in the airflow path through the enclosure space to cool the process fluid with the air drawn in from the external environment as it flows through the radiator assembly.

(10) The system's enclosure assembly may also include one or more outlet fan assemblies positioned in fluid communication with the external environment. In order to maintain a desired temperature of the enclosure space, the one or more outlet fan assemblies may be operatively connected to one or more outlet fan motors to discharge air from the enclosure space to the external environment when the one or more outlet fan motors is activated so that airflow in the enclosure space that has been heated from the cooling of the process fluid may be ventilated from the enclosure assembly. The enclosure assembly of the system may also include one or more temperature sensors to detect a temperature of the enclosure space and, further, one or more controllers in electrical communication with the one or more temperature sensors. The one or more controllers may be operatively connected to one or more of the one or more intake fan motors and the one or more outlet fan motors. In this regard, the one or more controllers may activate the respective one or more intake fan motors and the one or more outlet fan motors to rotate the respective one or more intake fan assemblies and the one or more outlet fan assemblies responsive to a predetermined temperature signal from the one or more temperature sensors to discharge heated air from and maintain a desired temperature of the enclosure space.

(11) Still another exemplary embodiment of the disclosure includes a method of enhancing cooling during operation of a hydraulic fracturing direct drive unit (DDU) having a turbine engine mechanically connected to a gearbox. The method may include operating the DDU to drive a driveshaft operatively connected to a fluid pump such that one or more of the turbine engines and the fluid pump generate and heat process fluid, for example, a lubrication or other lubrication medium, and/or a hydraulic/working fluid. The method may include detecting a temperature in an enclosure space of an enclosure assembly housing the DDU with one or more temperature sensors, and, further, controlling one or more intake fan assemblies of one or more heat exchanger assemblies in the enclosure space to draw air from an external environment into an airflow through the enclosure space based upon a temperature signal detected by the one or more temperature sensors. In this regard, the method may include cooling the process fluid by directing airflow from the one or more intake fan assemblies toward one or more radiator assemblies of the one or more heat exchangers carrying the process fluid. The method may further include controlling one or more outlet fan assemblies to discharge airflow heated by the cooling of the process fluid to the external environment to maintain a desired temperature in the enclosure space.

(12) Those skilled in the art will appreciate the benefits of various additional embodiments reading the following detailed description of the

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) According to common practice, the various features of the drawings discussed below are not necessarily drawn to scale. Dimensions of various features and elements in the drawings may be expanded or reduced to more clearly illustrate the embodiments of the disclosure.
- (2) FIG. 1A is a schematic diagram of a pumping unit according to an embodiment of the disclosure.
- (3) FIG. 1B is a schematic diagram of a layout of a fluid pumping system according to an embodiment of the disclosure.
- (4) FIG. 2 is a perspective view of an enclosure assembly according to an embodiment of the disclosure.
- (5) FIG. 3 is a schematic sectional view of an enclosure body according to an embodiment of the disclosure.
- (6) FIG. 4 is a schematic sectional view of an enclosure assembly according to an embodiment of the disclosure.
- (7) FIG. 5 is a schematic diagram of a heat exchanger assembly according to an embodiment of the disclosure.
- (8) FIG. 6 is a front view of a heat exchanger assembly according to an embodiment of the disclosure.
- (9) FIG. 7 is a side view of a heat exchanger assembly according to an embodiment of the disclosure.
- (10) FIG. 8 is a plan sectional view of an enclosure assembly according to an embodiment of the disclosure.
- (11) FIG. 9 is a side sectional view of an enclosure assembly according to an embodiment of the disclosure.
- (12) FIG. 10 is a schematic diagram of a hydraulic circuit according to an embodiment of the disclosure.
- (13) FIG. 11 is a schematic diagram of a control circuit according to an embodiment of the disclosure.
- (14) Corresponding parts are designated by corresponding reference numbers throughout the drawings.

DETAILED DESCRIPTION

(15) The embodiments of the present disclosure are directed to enclosure assemblies to enhance cooling of a hydraulic fracturing direct drive unit (DDU) during operation. The embodiments of the present disclosure may be directed to such enclosure assemblies for enhanced cooling of DDUs associated with high-pressure, high-power hydraulic fracturing operations.

(16) FIG. 1A illustrates a schematic view of a pumping unit **111** for use in a high-pressure, high power, fluid pumping system **113** (FIG. 1B) for use in hydraulic fracturing operations according to an embodiment of the disclosure. FIG. 1B shows a typical pad layout of the pumping units **111** (indicated as FP1, FP2, FP3, FP4, FP5, FP6, FP7, FP8) with the pumping units all operatively connected to a manifold M that is operatively connected to a wellhead W.

(17) By way of an example, the system **113** is a hydraulic fracturing application that may be sized to achieve a maximum rated horsepower of 24,000 HP for the pumping system **113**, including a quantity of eight (8) 3000 horsepower (HP) pumping units **111** that may be used in one embodiment of the disclosure. It will be understood that the fluid pumping system **113** may include associated service equipment such as hoses, connections, and assemblies, among other devices and tools. As shown in FIG. 1A, each of the pumping units **111** are mounted on a trailer **115** for transport and positioning at the jobsite. Each pumping unit **111** includes an enclosure assembly **121** that houses a direct drive unit (DDU) **123** including a gas turbine engine **125** operatively connected to a gearbox **127** or other mechanical transmission.

(18) The pumping unit **111** has a driveshaft **131** operatively connected to the gearbox **127**. The pumping unit **111** includes a high-pressure, high-power, reciprocating positive displacement pump **133** that is operatively connected to the DDU **123** via the driveshaft **131**. In one embodiment, the pumping unit **111** is mounted on the trailer **115** adjacent the DDU **123**.

(19) The trailer **115** includes other associated components such as a turbine exhaust duct **135** operatively connected to the gas turbine engine **125**, air intake duct **137** operatively connected to the gas turbine, and other associated equipment hoses, connections, or other components as will be understood by those skilled in the art to facilitate operation of the fluid pumping unit **111**.

(20) In the illustrated embodiment, the gas turbine engine **125** may be a Vericor Model TF50F bi-fuel turbine; however, the DDU **123** may include other gas turbines or suitable drive units, systems, and/or mechanisms suitable for use as a hydraulic fracturing pump drive without departing from the disclosure. In one embodiment, the fluid pumping system **113** may include a turbine engine that uses diesel or other fuel as a power source. The gas turbine engine **125** is cantilever mounted to the gearbox **127**, with the gearbox **127** supported by the floor of the enclosure assembly **121**.

(21) It should also be noted that, while the disclosure primarily describes the systems and mechanisms for use with DDUs **123** to operate fracturing pumping units **111**, the disclosed systems and mechanisms may also be directed to other equipment within the well stimulation industry such as, for example, blenders, cementing units, power generators and related equipment, without departing from the scope of the disclosure.

(22) FIGS. 2 and 4 illustrate an enclosure assembly **121** that houses the DDU **123** according to an exemplary embodiment of the disclosure. As shown, the enclosure assembly **121** includes an enclosure body **165** that may extend at least partially around an enclosure space **122** to house one or more portion of the DDU **123** therein. The enclosure space **122** may also be sized and configured to accommodate other DDU/engine equipment, for example, a driveshaft interface, fuel trains, an exhaust system flanged connection, a fire suppression system, bulkheads, exhaust ducting, engine air intake ducting, hydraulic/pneumatic bulkhead hoses, inspection doors/hatches, or other components and equipment as will be understood by those skilled in the art.

(23) In one embodiment, the enclosure body **165** may be a generally box-like or cuboid arrangement of walls, including a first side wall **167**, a second side wall **169** opposite the first side wall **167**, and an opposing front wall **171** and rear wall **173** each extending from the first side wall **167** to the second side wall **169**. The enclosure body **165** may also include a roof/top wall **166** (FIG. 4) and a floor/bottom wall **168**. In one embodiment, the floor **168** may be formed of a solid base steel material mounted on a skid structure.

(24) Referring additionally to FIG. 3, one or more of the walls of the enclosure body **165** may be provided with sound-attenuating, e.g., vibration-dampening, properties to minimize the transmission of sound from one or more operations of the DDU **123**, e.g., running of the turbine engine **125** and/or the gearbox **127**, from the enclosure space **122** to an external environment surrounding the enclosure body **165**. In this regard, the walls of the enclosure body **165** may have a configuration in which multiple layers are arranged to provide sound attenuation. Other sound-attenuating features may be incorporated into the construction of the enclosure assembly **121**. For example, the gearbox **127** may be provided with shock-absorbing feet or mounts that minimize the transmission of vibrations to the enclosure body **165**.

(25) In one embodiment, the walls of the enclosure body **165** may include an outer metallic layer **171**, a foam or other polymeric layer **173** and a composite layer **175**, and in inner or liner metallic layer **177**, with the foam layer **173** and the composite layer **175** positioned between the metallic layers **171**, **177**.

(26) In one embodiment, the walls **167**, **169**, **171**, **173** of the enclosure body **165** may be formed from approximately 12"×12" panels with an overall thickness of about 4.5" to about 5.25" that may clip, snap, or otherwise connect together in a generally modular arrangement, and the

outer metallic layer **171** may be, for example, a 22 ga perforated aluminum sheet, the foam layer **173** may be, for example, a 1" foam layer, the composite layer **175** may be, for example, a 3"-4" layer of mineral wool, and the inner metallic layer **177** may be, for example, perforated 22 ga aluminum. The roof **166** of the enclosure body **165** may have a similar arrangement, with an overall thickness of, for example, about 2" and having the foam layer **173** at a thickness of about, for example, 1.5". The enclosure body **165** may have a different arrangement without departing from the disclosure.

(27) Still referring to FIG. 2, a plurality of doors may be movably connected/attached to the enclosure body **165**, e.g., to provide access to the enclosure space **122** for inspections, maintenance, or other operations as will be understood by those skilled in the art. A pair of doors **179** may be hingably connected/attached to the first side wall **167** of the enclosure body **165** to provide access to the enclosure space **122** through openings formed in the first side wall **167** upon movement of the doors **179**.

(28) A door **181** may also be movably connected to the second side wall **169** of the enclosure body **165** to provide access to the enclosure space **122** along the second side wall **169**. In one embodiment, the door **181** may be slidably connected/attached to the second side wall **169** on rails, tracks, or other guides as will be understood by those skilled in the art, such that slidable movement of the door **181** exposes an opening in the second side wall **169** through which an operator may access the enclosure space **122**. In one embodiment, the door **181** may have one or more foldable or otherwise reconfigurable portions.

(29) With additional reference to FIG. 4, a generally horizontal partition **183** may extend in general parallel relation with the roof **166** and the floor **168** of the enclosure body **165** so as to provide an upper compartment **185** and a lower compartment **187** of the enclosure space **122**. In one embodiment, the upper compartment **185** may include an air intake assembly that may include an arrangement of ducts, fans, ports, filtration assemblies, blowers, compressors, cooling coils, or other components as will be understood by those skilled in the art, to feed filtered air into the turbine engine **123** positioned in the lower compartment **187**.

(30) In view of the foregoing, the enclosure assembly **121** may be provided with a generally weatherproof or weather-resistant configuration that is sufficiently robust for use in hydraulic fracturing applications, and which additionally provides sound attenuation properties for enclosed and associated equipment. For example, the enclosure assembly **121** may provide sufficient sound attenuation emanating from one or more incorporated heat exchanger assemblies, as described further herein.

(31) During various operations of the pumping unit **133**, e.g., startup and shutdown procedures, idling, maintenance cycles, active driving of the pumping unit **133**, or other operations as will be understood by those skilled in the art, heat may be generated in one or more portions of the pumping unit **133**, for example, via frictional engagement of components of the pumping unit **133** such as pistons, bores, or other components as will be understood by those skilled in the art. In this regard, the pumping unit **133** may employ a fluid heat transfer medium, e.g., a natural or synthetic lubrication oil, to absorb heat from the pumping unit **133** via fluid convection to reduce heat in one or more portions of the DDU **123**.

(32) Similarly, during various operations of the DDU **123**, heat may be generated by one or more portions of the turbine engine **125** and the gearbox **127**. The DDU **123** may thus also employ a fluid heat transfer medium to absorb heat from the DDU **123** via fluid convection to reduce heat in one or more portions of the DDU **123**.

(33) Further, various hydraulic components of the fluid pumping system **113**, e.g., actuators, motors, pumps, blowers, coolers, filters, or other hydraulic components as will be understood by those skilled in the art, that receive pressurized hydraulic fluid or working fluid therethrough may cause such hydraulic fluid/working fluid to increase in temperature during the course of such operation.

(34) The aforementioned fluid heat transfer media, hydraulic fluids/working fluids, and other thermally conductive fluids associated with the fluid pumping system **113** may be collectively referred to as process fluids associated with the respective components of the fluid pumping system **113** herein.

(35) In this regard, the fluid pumping system **113** may include one or more heat exchanger assemblies for cooling/reducing heat in the aforementioned process fluids. Turning to FIG. 5, a heat exchanger assembly **189A** according to an exemplary embodiment of the disclosure is schematically illustrated. In the illustrated embodiment, the heat exchanger assembly **189A** may be connected to, e.g., attached, mounted, or otherwise supported by, the enclosure body **165**. While the heat exchanger assembly **189A** is illustrated as being positioned in the enclosure space **122**, it will be understood that the heat exchanger assembly **189A** may be connected to the enclosure body **165** and at least partially positioned outside thereof without departing from the disclosure.

(36) Still referring to FIG. 5, the heat exchanger assembly **189A** may include one or more intake fan assemblies **193**, one or more intake fan motors **195** operatively connected to the intake fan assembly **193**, and one or more radiator assemblies **197** positioned adjacent the intake fan assembly **193**. The heat exchanger assembly **189A** may be positioned in alignment with a cutout or opening in the enclosure body **122**, e.g., so that the heat exchanger assembly **189A** may be in at least partial fluid communication with an external environment **E** surrounding the enclosure assembly **121**. In one embodiment, such cutout or opening may be at least partially covered with a mounting plate **194** which may be connected to the heat exchanger assembly **189A**.

(37) A sealing member **198**, for example, a gasket or other polymeric member, may be positioned between the heat exchanger assembly **189A** and the enclosure body **165**, for example, to inhibit the migration or leakage of fluids between the heat exchanger assembly **189A** and the enclosure body **165**.

(38) The one or more intake fan assemblies **193** may include one or more fans **205** (FIG. 6) rotatably connected to the intake fan motor **195** such that, upon receiving a driving signal or other modality of actuation, the intake fan motor **195** rotates the one or more fans **205** to rotate and circulate air through the enclosure space **122**. Such rotatable connection between the intake fan motor **195** and the fan **205** may be a driveshaft, coupling, or other mechanical transmission. The fan **205** may have a plurality of blades/arms for forcing/urging air into an airflow. In this regard, the fan **205** may be provided with blades/arms having a length, pitch, shape, or other features as will be understood by those skilled in the art, configured to influence airflow in a preselected direction.

(39) As shown, the one or more radiator assemblies **197** is positioned adjacent the intake fan assembly **193**. In one embodiment, the radiator assembly **197** may be configured as a tube-and-shell heat exchanger, in which one or more conduits (e.g., tubes, ducts, hoses, fluid lines, or other conduits as will be understood by those skilled in the art) extend along bulkhead fittings on the enclosure body **122** and through an interior of a housing shell **207** to route the process fluid over a sufficient surface area to effect cooling of the process fluid.

(40) The conduits extending through the housing shell **207** may carry process fluid in the form of a fluid heat exchange medium, hydraulic fluid/working fluid, or other fluid. As described further herein, the radiator assembly **197** may be positioned in an airflow path at least partially provided by the intake fan assembly **193** to remove heat from the process fluid running through the conduits. In one embodiment, the radiator assembly **197** may be covered by/positioned adjacent one or more layers of mesh or otherwise porous material.

(41) Referring to FIGS. 6 and 7, the enclosure assembly **121** may include the heat exchanger assembly **189A** (broadly, "low-pressure heat exchanger assembly **189A**") for cooling process fluid received from a low-pressure portion of the fluid pump **133**, and the enclosure assembly **121** may further include a high-pressure heat exchanger assembly **189B** for cooling process fluid received from a high-pressure portion of the fluid pump **133**. The heat exchanger assembly **189B** may be similarly configured to the heat exchanger assembly **189A**, though the heat

exchanger assemblies **189A**, **189B** may have one or more differences without departing from the disclosure.

(42) As also shown, the heat exchanger assemblies **189A**, **189B** are supported on a mounting frame **191** with a generally rigid body having outer frame members **199**, **200** intersecting at respective joints/plates **201** that may be secured with fasteners such as bolts, screws, rivets, pins, or other fasteners as will be understood by those skilled in the art. As also shown, the mounting frame **191** is provided with one or more flanges or securing tabs **203** extending from one or more of the frame members **199**, **200** and that are configured for engagement with the enclosure body **165**. In this regard, the securing tabs **203** may have, for example, a generally flat or planar profile and/or may be provided with an opening for receiving a fastener therethrough. In one embodiment, the securing tabs **203** may be integrally formed with one or more of the frame members **199**, **200**.

(43) The heat exchanger assemblies **189A**, **189B** may both be connected to the mounting frame **191** in a vertically stacked arrangement, as shown, though each heat exchanger assembly **189A**, **189B** may be connected to the enclosure body **165** on separate mounting frames without departing from the disclosure. In one embodiment, the mounting frame **191** may be about 0.25" thick, and may be provided with a tolerance of about 0.1" to about 0.2" beyond the boundaries of the heat exchanger assemblies **189A**, **189B**.

(44) In one embodiment, the mounting frame **191** may be connected to a modular panel of the side wall **167** that is sized and configured to an area larger than that of the heat exchanger assemblies **189A**, **189B**. In one embodiment, such modular panel may be provided with a tolerance of about 0.35" to about 0.45" beyond the heat exchanger assemblies **189A**, **189B**.

(45) In one embodiment, and as shown in FIG. 2, the enclosure assembly **121** may include additional or alternative heat exchangers, for example, a heat exchanger **189C** for cooling process fluid associated with the turbine engine **125**, a heat exchanger **189D** for cooling process fluid associated with the gearbox **127**, and a heat exchanger **189E** for cooling process fluid associated with one or more hydraulic components of the fluid pumping system **113** (e.g., auxiliary/ancillary actuators, pumps, motors, or other hydraulic components as will be understood by those skilled in the art). It will be understood that each of the heat exchanger assemblies **189A**, **189B**, **189C**, **189D**, **189E** may be sized/scaled/configured according to the process fluids upon which they are operative to cool.

(46) As described herein, the heat exchanger assemblies **189C**, **189D**, **189E** may have a configuration that is substantially similar to that of the heat exchanger assemblies **189A** and **189B**, though one or more of the heat exchanger assemblies **189A**, **189B**, **189C**, **189D**, **189E** may have a different configuration without departing from the disclosure. By way of example, two or more of the one or more of the heat exchanger assemblies **189A**, **189B**, **189C**, **189D**, **189E** may share a common mounting frame, housing shell, intake fan assembly, or other component as will be understood by those skilled in the art.

(47) As shown in FIG. 2, the heat exchanger assemblies **189A**, **189B**, **189C**, **189D**, **189E** are connected to the enclosure body **165** and positioned in fluid communication with the external environment E such that when the respective intake fan assemblies **193** are driven by the respective intake fan motors **195**, the intake fan assemblies **193** are operative to draw air in from the external environment E toward the respective radiator assemblies **197** to remove heat/cool the process fluids flowing therethrough, and so that they may return to respective portions of the fluid pumping system for continued lubrication/cooling of components of the fluid pumping system **113**.

(48) The aforementioned action of the intake fan assemblies **193** causes air from the external environment E to absorb heat from the radiator assemblies **197** as it passes thereby/therethrough and further into the enclosure space **122**. In this regard, operation of one or more of the heat exchanger assemblies **189A**, **189B**, **189C**, **189D**, **189E** may cause an ambient temperature in the enclosure space **122** of the enclosure assembly **121** to increase.

(49) With additional reference to FIGS. 8 and 9, one or more outlet/suction fan assemblies **209** may also be connected to the enclosure body **165**. The one or more outlet fan assemblies **209** may have a similar configuration to the aforementioned intake fan assemblies **193**, in that they may include one or more outlet fans, e.g., a fan **205**, in operative communication with one or more respective motors, e.g., an outlet fan motor **196**, such that upon receiving a driving signal or actuation force, the outlet fan motor **196** may drive the fan **205** to rotate. In one embodiment, the outlet fan assembly **209** may include a pair of fans **205** driven by one or more outlet fan motors **196**. It will be understood that the one or more inlet fan assemblies **193** and the one or more outlet fan assemblies **209** may be driven by the same motor or combination of motors. Although the one or more outlet fan assemblies **209** has been described herein separately from the heat exchanger assemblies **189A**, **189B**, **189C**, **189D**, **189E**, it will be understood that one or more of the heat exchanger assemblies **189A**, **189B**, **189C**, **189D**, **189E** may include the one or more outlet fan assemblies **209** without departing from the disclosure.

(50) In one embodiment, one or more of the heat exchanger assemblies **189A**, **189B**, **189C**, **189D**, **189E** may be attached to the first side wall **167** of the enclosure body **165**, and the outlet fan assembly **209** may be attached to the second side wall **169** of the enclosure body **165**. It will be understood that the heat exchanger assemblies **189A**, **189B**, **189C**, **189D**, **189E** and the outlet fan assembly **209** may be attached to the enclosure body **165** in a different arrangement without departing from the disclosure.

(51) In this regard, upon receipt of an actuation force or driving signal, the one or more outlet fan motors **196** associated with the outlet fan assembly **209** may rotate the fan **205** to discharge air from the enclosure space **122** to the external environment E. Accordingly, the arrangement of the heat exchanger assemblies **189A**, **189B**, **189C**, **189D**, **189E** and the outlet fan assembly **209** is operative to draw atmospheric/cool air in from the external environment E at the intake fan assembly **193**, direct airflow toward the radiator assembly **197** to cool the process fluids flowing therethrough, and, further, to ventilate the enclosure assembly **121** by directing an airflow path A from the intake fan assembly **193** to the outlet fan assembly **209** and discharging the air from the enclosure space **122**/airflow path A that has been heated from cooling the radiator assembly **197** to the external environment E at the outlet fan assembly **209**.

(52) Still referring to FIGS. 8 and 9, in one embodiment, one or more of the heat exchanger assemblies **189A**, **189B**, **189C**, **189D**, **189E**, in cooperation with the one or more outlet fan assemblies **209**, is configured to replace a volume of air in the enclosure space **122** at an interval of about 30 seconds. It will be understood that the heat exchanger assemblies may be configured to replace the same or a different volume of air at a different time interval without departing from the disclosure.

(53) Accordingly, the enclosure assembly **121** may be provided with enhanced cooling capabilities for managing excess heat generated by one or more of the DDU **123**, the fluid pump **113**, and various hydraulic components associated with the fluid pumping system **113**. As described above, one or more of the heat exchanger assemblies **189A**, **189B**, **189C**, **189D**, **189E** is operative to cool process fluid associated with one or more of the DDU **123**, the fluid pump **113**, and various hydraulic components associated with the fluid pumping system **113**. Further, the intake fan assemblies **193** of the heat exchanger assemblies **189A**, **189B**, **189C**, **189D**, **189E** direct the airflow path A through the enclosure space **122** such that, in cooperation with the outlet fan assembly **209**, the air in the enclosure space **122** may be discharged to the external environment E to provide ventilation in the enclosure space **122**. Such ventilation may, for example, maintain a desired temperature of the enclosure space **122**, e.g., to further enhance a temperature differential between the airflow path A and the process fluid in the heat exchanger assemblies **189A**, **189B**, **189C**, **189D**, **189E**.

(54) As described herein, one or more of the motors **195**, **196** may be hydraulic motors, e.g., such that a pressurized working fluid/hydraulic fluid flows therethrough to actuate the motors **195**, **196**.

(55) With additional reference to FIG. 10, a schematic diagram is provided to show a hydraulic circuit that may be used to drive one or more

of the fans **205** of the respective intake fan assemblies **193**. As shown, each intake fan motor **195** includes an inlet port **211** in fluid communication with a hydraulic pump **213** to receive pressurized fluid from the hydraulic pump **213** to actuate the respective intake fan motor **195**. The intake fan motors **195** are also in fluid communication with a return port or outlet port **215** in fluid communication with the hydraulic pump **213** to return hydraulic fluid/working fluid to the respective hydraulic pump **213** after it has passed through/actuated the respective intake fan motor **195**. Each intake fan motor **195** may also include a drain port **217** in fluid communication therewith, for example, to provide drainage of overflow/excess hydraulic fluid/working fluid, to provide a leakage path or pressure release, or other fluid release as will be understood by those skilled in the art. It will be understood that the one or more outlet fan motors **196** may be arranged/controlled in a manner similar to that described above with regard to the inlet fan motors **195**.

(56) It will be understood that the hydraulic pump **213** may be in fluid communication with the respective fluid pump **133**, the turbine engine **125**, the gearbox **127**, and one or more hydraulic components of the fluid pumping system **113** to receive and return process fluid thereto, for example, through an arrangement of fluid lines, manifolds, valves, or other fluid conduit as will be understood by those skilled in the art. In one embodiment, each of the fluid pump **133**, the turbine engine **125**, the gearbox **127**, and one or more hydraulic components of the fluid pumping system **113** may be associated with a separate hydraulic pump **213**, or a combination of hydraulic pumps **213**. In one embodiment, the motors **195** associated with the respective low-pressure portion of the fluid pump **133** and the high-pressure portion of the fluid pump **133** may share one or more common fluid lines.

(57) Each intake fan motor **195** may have an associated solenoid **219** that includes one or more fluid valves to control the flow of hydraulic fluid/working fluid thereto and therefrom. For example, upon receipt of a predetermined electrical signal, each solenoid **219** may actuate, e.g., open or dilate, to permit the flow of hydraulic fluid/working fluid from the hydraulic pump **213** to the respective inlet port **211** and to permit the flow of hydraulic fluid/working fluid from the respective outlet portion **215** to the hydraulic pump **213**. Similarly, the solenoid **219** may close, e.g., restrict or block, the flow of hydraulic fluid/working fluid therethrough upon receipt of a predetermined electrical signal, e.g., a closure signal.

(58) While the intake fan motors **195** described herein have been described as hydraulic motors driven by pressurized hydraulic/working fluid, it will be understood that one or more of the motors **195** (or the motors **196**) may be an electric motor driven by a received electrical actuation/driving signal. In one embodiment, one or more of the motors **195**, **196** may be an electric motor powered from 3-phase electrical power provided by an onboard generator system capable of a voltage output of 480V.

(59) Turning to FIG. **11**, a schematic diagram of a control system that may be used to control the inlet fan motors **195** is illustrated. As shown, each solenoid **219** may be electrically connected to a controller **221**, e.g., a programmable logic controller (PLC), an off-highway multi-controller, a processor-implemented controller, or other control feature as will be understood by those skilled in the art. In this regard, the controller **221** may be operable to actuate the solenoids **219**, e.g., to selectively open and close the valves of the solenoid **219** to permit/restrict the flow of hydraulic fluid/working fluid through the respective inlet fan motors **195**. It will be understood that the one or more outlet fan motors **196** may be controlled in a manner similar to that described above with regard to the inlet fan motors **195**.

(60) In this regard, the controller **221** may be configured to transmit a driving or actuation signal to the respective solenoids **219** upon receipt of a predetermined electrical signal from a thermal/temperature sensor **223** that may be in proximity to the process fluid associated with the respective turbine engine **125**, gearbox **127**, low-pressure portion of the pump **133**, the high-pressure portion of the pump **133**, and one or more hydraulic components of the fluid pumping system **113**. In this regard, one or more temperature sensors **223** may be connected to the enclosure assembly **121** or components thereof. In one embodiment, the sensors **223** may be disposed along a fluid line between the outlet port/return portion **215** of the respective motor **195** and the hydraulic pump **213** and/or a respective reservoir for the process fluid carried therethrough.

(61) In one embodiment, the sensors **223** may be digital thermometers or another electronic sensor that may receive/absorb heat from the associated respective turbine engine **125**, gearbox **127**, low-pressure portion of the pump **133**, the high-pressure portion of the pump **133**, and one or more hydraulic components of the fluid pumping system **113**, and transmit a corresponding electrical signal to the controller **221**. If the respective electrical signal corresponds to a temperature that is at or above a predetermined value or threshold, for example, set by regulation or OEMs, the controller **221** may signal the respective solenoid **219** to open the respective valves.

(62) It will be understood that such actuation of the solenoids **219** may be performed at a constant or predetermined time interval, on-demand, e.g., if and when a predetermined signal is received from the sensors **223**, and/or may be performed proportionally to the temperature of the enclosure space **122**, e.g., so that determining and monitoring greater/lesser temperatures in the enclosure space **122**, the controller **221** will proportionally increase/decrease the flow rate of hydraulic/working fluid flowing through the respective intake fan motors **195**, and consequently, the speed of the respective associated fans **205**.

(63) In one embodiment, one or more of the sensors **223** may include an analog device configured to receive/absorb heat and product a corresponding analog electrical signal without any intermediate processing steps, for example, as in a thermocouple, resistance temperature detector (RTD), or temperature switch. Such analog electrical signal may be a raw value determined by the controller **221** or other processor to correspond to a temperature of the enclosure space **122**.

(64) While the hydraulic circuit and control of the respective fans **205** has been described above with regard to the heat exchanger assemblies **189A**, **189B**, **189C**, **189D**, **189E**, it will be understood that the fans **205** of the outlet fan assembly **209** may be driven and controlled in the same or a similar manner.

(65) Still other embodiments of the disclosure, as shown in FIGS. **1-11**, also include methods of enhancing cooling during operation of a hydraulic fracturing direct drive unit (DDU) having a turbine engine mechanically connected to a gearbox. An embodiment of a method may include operating the DDU to drive a driveshaft operatively connected to a fluid pump such that one or more of the turbine engine and the fluid pump generates and heats process fluid, for example, a lubrication or other lubrication medium, and/or a hydraulic/working fluid. The method may include detecting a temperature in an enclosure space of an enclosure assembly housing the DDU with one or more temperature sensors, and, further, controlling one or more intake fan assemblies of one or more heat exchanger assemblies in the enclosure space to draw air from an external environment into an airflow through the enclosure space based upon a temperature signal detected by the one or more temperature sensors. In this regard, the method may include cooling the process fluid by directing airflow from the one or more intake fan assemblies toward one or more radiator assemblies of the one or more heat exchangers carrying the process fluid. The method may further include controlling one or more outlet fan assemblies to discharge airflow heated by the cooling of the process fluid to the external environment to maintain a desired temperature in the enclosure space.

(66) In view of the foregoing, the disclosed embodiments of enclosure assemblies for DDUs may provide for enhanced cooling by the configuration and arrangement of one or more heat exchangers that cool one or more process fluids associated with the DDU and/or an associated fluid pumping system while also providing ventilation and cooling of an enclosure space within the enclosure assembly. In addition to the enhanced cooling of the DDU provided by such an arrangement, the footprint of the enclosure assembly may be minimized and the management of associated power systems may be streamlined.

(67) This is a continuation of U.S. Non-Provisional application Ser. No. 18/096,927, filed Jan. 13, 2023, titled “ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT AND RELATED METHODS,” which is a continuation of U.S. Non-Provisional application Ser. No. 17/444,485, filed Aug. 5, 2021, titled “ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT AND RELATED METHODS,” now U.S. Pat. No. 11,627,683, issued Apr. 11, 2023, which is a continuation of U.S. Non-Provisional application Ser. No. 17/356,063, filed Jun. 23, 2021, titled “ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT AND RELATED METHODS,” now U.S. Pat. No. 11,129,295, issued Sep. 21, 2021, which is a divisional of U.S. Non-Provisional application Ser. No. 17/302,039, filed Apr. 22, 2021, titled “ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT AND RELATED METHODS,” now U.S. Pat. No. 11,109,508, issued Aug. 31, 2021, which claims priority to and the benefit of, under 35 U.S.C. § 119(e), U.S. Provisional Application No. 62/705,042, filed Jun. 9, 2020, titled “ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT AND RELATED METHODS,” and U.S. Provisional Application No. 62/704,981, filed Jun. 5, 2020, titled “ENCLOSURE ASSEMBLY FOR ENHANCED COOLING OF DIRECT DRIVE UNIT (DDU) AND RELATED METHODS,” the disclosures of which are incorporated herein by reference in their entireties.

(68) The foregoing description of the disclosure illustrates and describes various exemplary embodiments. Various additions, modifications, and changes may be made to the exemplary embodiments without departing from the spirit and scope of the disclosure. It is intended that all matter contained in the above description or shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense. Additionally, the disclosure shows and describes only selected embodiments of the disclosure, but the disclosure is capable of use in various other combinations, modifications, and environments and is capable of changes or modifications within the scope of the inventive concept as expressed herein, commensurate with the above teachings, and/or within the skill or knowledge of the relevant art. Furthermore, certain features and characteristics of each embodiment may be selectively interchanged and applied to other illustrated and non-illustrated embodiments of the disclosure.

Claims

1. An enclosure assembly of a hydraulic fracturing power drive unit, the enclosure assembly comprising: an enclosure body at least partially defining an enclosure space to house the hydraulic fracturing power drive unit, the enclosure body comprising: a floor; a plurality of side walls; wherein each of the side walls comprises a plurality of layers, and wherein at least one of the plurality of layers is configured to restrict sound transmission therethrough; wherein the plurality of layers includes: an inner metallic layer; an outer metallic layer; and a foam layer and a wool layer positioned between the inner metallic layer and the outer metallic layer; a roof; and a pair of doors hingably attached to a first side wall of the plurality of side walls; one or more fan assemblies configured to flow air from an external environment surrounding the enclosure body along an airflow path through the enclosure space; a temperature sensor configured to detect a temperature of the enclosure space and output a temperature signal; and a controller in electrical communication with the temperature sensor, the controller configured to activate at least one of the one or more fan assemblies responsive to the temperature signal, thereby to maintain a desired temperature of the enclosure space.
2. The enclosure assembly of claim 1, wherein the one or more fan assemblies includes an intake fan assembly connected to one of the plurality of side walls, the intake fan assembly comprising a motor.
3. The enclosure assembly of claim 1, wherein the one or more fan assemblies includes an outtake fan assembly connected to one of the plurality of side walls, the outtake fan assembly comprising a motor.
4. The enclosure assembly of claim 1, wherein the one or more fan assemblies includes an intake fan assembly and an outtake fan assembly, wherein the intake fan assembly is connected to a side wall of the enclosure body and the outtake fan assembly is connected to another, separate side wall of the enclosure body such that the airflow path extends from the intake fan assembly to the outtake fan assembly within the enclosure space.
5. The enclosure assembly of claim 1, wherein the temperature sensor includes: (a) an analog sensor configured to produce an analog electrical signal corresponding to the temperature of the enclosure space, or (b) a digital sensor configured to produce a digital electrical signal corresponding to the temperature of the enclosure space.
6. The enclosure assembly of claim 1, wherein one of the plurality of side walls is configured to receive a driveshaft therethrough.
7. The enclosure assembly of claim 6, wherein the driveshaft is connected to a gearbox disposed within the enclosure body, the gearbox having shock-absorbing mounts configured to reduce transmission of vibrations from the gearbox to the enclosure body.
8. The enclosure assembly of claim 7, further comprising: a gas turbine cantilever mounted to the gearbox and disposed within the enclosure body; and an air intake assembly disposed in the enclosure body in a location between the gas turbine and the roof.
9. A fluid pumping system for hydraulic fracturing operations, the fluid pumping system comprising: a power drive unit, comprising: a gearbox; a gas turbine engine cantilever mounted to the gearbox; and a driveshaft operably connected to the gearbox; a fluid pump operatively connected to the driveshaft; an enclosure assembly including an enclosure body at least partially defining an enclosure space to house at least a portion of the power drive unit, the enclosure body comprising: a floor; a plurality of side walls; wherein each of the side walls comprises a plurality of layers, and wherein at least one of the plurality of layers is configured to restrict sound transmission therethrough; wherein the plurality of layers includes: an inner metallic layer; an outer metallic layer; and a foam layer and a wool layer positioned between the inner metallic layer and the outer metallic layer; a roof; and a pair of doors hingably attached to a first side wall of the plurality of side walls; one or more fan assemblies configured to flow air from an external environment surrounding the enclosure body along an airflow path through the enclosure space; and a temperature sensor configured to detect a temperature of the enclosure space and configured to output a temperature signal.
10. The fluid pumping system of claim 9, wherein the one or more fan assemblies includes an intake fan assembly or an outtake fan assembly, wherein the intake fan assembly is connected to a side wall of the enclosure body and the outtake fan assembly is connected to another, separate side wall of the enclosure body such that the airflow path extends from the intake fan assembly to the outtake fan assembly within the enclosure space.
11. The fluid pumping system of claim 9, wherein one of the plurality of side walls is configured to receive the driveshaft therethrough.
12. The fluid pumping system of claim 11, wherein the gearbox is disposed upon shock-absorbing mounts configured to reduce transmission of vibrations from the gearbox to the enclosure body.
13. The fluid pumping system of claim 11, wherein the gearbox is supported by the floor of the enclosure body.
14. A fluid pumping system for hydraulic fracturing operations, the fluid pumping system comprising: a power drive unit, comprising: a gearbox; a gas turbine engine cantilever mounted to the gearbox; and a driveshaft operably connected to the gearbox; a fluid pump operatively connected to the driveshaft; an enclosure assembly including an enclosure body at least partially defining an enclosure space to house at least a portion of the power drive unit, the enclosure body comprising: a floor; a plurality of side walls; wherein each of the side

walls comprises a plurality of layers, and wherein at least one of the plurality of layers is configured to restrict sound transmission therethrough; wherein the plurality of layers includes: an inner metallic layer; an outer metallic layer; and a foam layer and a wool layer positioned between the inner metallic layer and the outer metallic layer; a roof; and a pair of doors hingably attached to a first side wall of the plurality of side walls; and one or more fan assemblies configured to flow air from an external environment surrounding the enclosure body along an airflow path through the enclosure space, wherein the power drive unit, the fluid pump, and the enclosure assembly are mounted on a trailer along a longitudinal axis of the driveshaft.

15. The fluid pumping system of claim 14, wherein the gearbox is supported by the floor of the enclosure body.

16. The fluid pumping system of claim 14, wherein the gearbox is disposed upon shock-absorbing mounts configured to reduce transmission of vibrations from the gearbox to the enclosure body.
